Medial-axis reconstruction on complete Riemannian manifolds: capture, drift, and the medial complex*

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Abstract

Let X be a nonempty proper closed subset of a complete Riemannian manifold M. The *two-geodesic medial axis* M_X^g is the set of points admitting at least two minimising geodesics to X — the multi-geodesic part of the cut locus of X — and the associated *reconstruction* $\mathcal{R}_X^g$ is the union of the open free balls $B(a, d_X(a))$ over $a \in M_X^g$. We characterise $\mathcal{R}_X^g$ on every complete M, with no curvature, genericity, analyticity, spread, or cardinality hypothesis. Two mechanisms drive the theory. A *capture theorem* shows that along any solution of the differential inclusion $\Phi' \in -\operatorname{conv} U(\Phi)$, driven by the unit directions of minimising geodesics, the distance d_X grows at unit rate until the first approach to M_X^g; hence every point p either lies at depth arbitrarily close to $d_X(p)$ inside medial balls, or its trajectory escapes to infinity. In particular $\mathcal{R}_X^g = M \setminus X$ on every compact manifold, for every closed X — even a single point — answering a question of Białożyt. Second, normalising free balls by their reach, the locally uniform limits $\psi = \lim[d(\cdot, a_k) - d_X(a_k)]$ over medial centres — *medial limit functions* — satisfy $\{\psi < 0\} \subseteq \mathcal{R}_X^g$, and $\mathcal{R}_X^g$ is exactly the union of these sublevel sets; every point is captured, drift-swallowed, or escaping. Białożyt's Euclidean characterisation, and the compact, spherical, hyperbolic, and Hadamard pictures, are specialisations; on the flat cylinder we prove a sharp end-certificate dichotomy. A topological layer follows: the escape class coincides with the Aubry set of d_X in the sense of Cannarsa–Cheng–Fathi, so the captured part of the complement always has the homotopy type of M_X^g, and when $\sup d_X < \infty$ the complement, the axis, and the cut locus are homotopy equivalent — uniform capture recovers the Ford–Voronoi spines of cusped hyperbolic manifolds and the Voronoi spines of crystallographic quotients, and for smooth X the captured set is the mapping cylinder of the swallow map onto the cut locus. Finally, every swallowed-uncaptured point lies in an *ideal* free horoball at depth no smaller than its depth in any finite medial ball; consequently the *medial complex* — M_X^g with the ideal free horoballs glued along the pattern of their overlaps — always carries the homotopy type of $\mathcal{R}_X^g$. Complete reconstruction does not transport homotopy type, already for three points in the plane; the medial complex repairs the leak unconditionally.

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*Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the two mechanisms of the capture theorem — the rate-one identity off the axis and the margin at first approach — were verified numerically on $\mathbb{S}^2$ against a non-circular convex contact curve whose focal region forces the foot to slide (60 random starts: growth rate 1 to five digits, margin deficit $< 3 \cdot 10^{-6}$; ancillary file `medial_capture_check.py`). An earlier version of parts of this material circulated as a series of informal notes. Remaining errors are ours.

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1 Introduction

1.1 The reconstruction problem

Fix a nonempty closed set X in a metric space M and, for each point p outside X, ask for the nearest points of X. Where the answer is unique the geometry of X is locally transparent; the interesting locus is the set of p with an *ambiguous* nearest-point answer. In the plane this is the classical *medial axis* or *skeleton*, a one-dimensional graph that threads the region between the features of X and has been used, since Blum, as a shape descriptor in vision, in mesh generation, and in the topology of domains [20, 11, 13]. Two features of the axis make it a genuine *descriptor* rather than a mere by-product: each axis point a carries a scalar, the distance $d_X(a)$ to X, and the open ball $B(a, d_X(a))$ of that radius is a *free ball* — a maximal ball that meets X only in its boundary. The union of the free balls over the axis,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B\big(a, d_X(a)\big),$$

is the *reconstruction* of the complement of X from the axis.

One wants to know when the reconstruction is faithful, i.e. when $\mathcal{R}_X = M \setminus X$. The obstruction is easy to see and hard to characterise. If X is convex in $\mathbb{R}^n$ the axis is empty and $\mathcal{R}_X = \varnothing$: convex sets cast no medial shadow, so their complements are not reconstructed at all. If X is a pair of points in the plane, the axis is the perpendicular bisector and its free balls reconstruct almost everything: the central ball already swallows the open segment between the points, and balls centred ever farther out along the bisector fill the two open half-planes bounded by the line through the pair — the deeper the point, the more remote the ball that reaches it, and the organising object is the *limit* of the escaping balls, an open half-plane. What no ball ever reaches are the two collinear rays beyond the points. The interplay between the local balls, which fill the "inside" of the hull, and the escaping balls, whose limits fill the outer half-spaces of defective faces of the hull, is the whole content of the Euclidean theorem.

1.2 Białożyt's theorem, the starting point

Białożyt [6] settled the Euclidean case completely. Write $\operatorname{conv} X$ for the convex hull and $\overline{\operatorname{conv}}\, X$ for its closure. A supporting hyperplane L of $\overline{\operatorname{conv}}\, X$ is *defective* if $X \cap L \neq \overline{\operatorname{conv}}\, X \cap L$, i.e. if X fails to fill the contact face that the hull presents on L; for each supporting $L = \{\langle v, \cdot\rangle = c\}$ (with $X \subseteq \{\langle v, \cdot\rangle \leq c\}$) let $L^+ = \{\langle v, \cdot\rangle > c\}$ be the open outer half-space.

Theorem 1.1 (Białożyt [6, Thm. 6])**.** *Let $X \subseteq \mathbb{R}^n$ be closed and nonempty. Then X is reconstructible, i.e. $\mathcal{R}_X = \mathbb{R}^n \setminus X$, if and only if*

$$\mathbb{R}^n \setminus X \;\subseteq\; \overline{\operatorname{conv}}\, X \;\cup \bigcup_{L \text{ defective}} L^+.$$

Two features of this statement organise everything that follows. First, the hull $\overline{\operatorname{conv}}\, X$ is reconstructed by *ordinary* (finite) medial balls: every point of $\overline{\operatorname{conv}}\, X \setminus X$ lies in a free ball centred on the axis [6]. Second, the outer half-spaces L^+ of defective supporting hyperplanes are reconstructed only in the *limit*, by free balls whose centres escape to infinity along a direction normal to L; a supporting hyperplane contributes such a limit precisely when its contact set is defective — equivalently, by Motzkin's theorem [23], when a point of L just outside the contact face is non-Chebyshev. This is the dichotomy *local balls vs. escaping balls* in its cleanest form, and the first goal of this paper is to find its counterpart on a curved manifold, where "half-space" must become "horoball", "convex hull" must become something curvature-dependent, and the very notion of "two nearest points" must be re-examined because a single nearest point can be reached by several competing geodesics. The second goal is the question Białożyt's theorem raises but cannot ask in $\mathbb{R}^n$: what happens on a *compact* manifold, where there is no infinity for the centres to escape to — a question posed explicitly in [6, §5.2]. The third goal, occupying the second half of the paper, is topological: when, and in what form, the axis carries the *homotopy type* of what it reconstructs.

1.3 From $\mathbb{R}^n$ to a manifold: what breaks

Let M be a complete Riemannian manifold. Three Euclidean conveniences fail at once.

(i) Geodesics are not unique. Two points can be joined by several minimising geodesics, and the nearest-point map to X can be single-valued (a unique foot $y \in X$) yet ambiguous as a *geodesic* problem (several minimisers reaching that same y). The sphere makes this vivid: for X a single point, every other point has that point as its unique nearest point, yet from the antipode it is reached by a whole sphere's worth of minimising geodesics — one nearest point, many nearest geodesics. This forces a distinction, invisible in $\mathbb{R}^n$, between the *two-point* axis M_X (two nearest *points*) and the *two-geodesic* axis M_X^g (two minimising *geodesics*); $M_X \subseteq M_X^g$, and it is M_X^g — the multi-geodesic part of the cut locus of X — that carries the reconstruction theory.

(ii) Convexity is not linear. There is no "conv X"; its role is played by different objects in different curvatures — the ordinary hull only in the flat case, a horoconvex hull in nonpositive curvature, and *nothing at all* on a compact manifold, where every complement turns out to be reconstructed and the hull condition is vacuous.

(iii) The distance function turns. In $\mathbb{R}^n$ the ascending gradient flow of d_X runs along straight rays until it meets the bisector; on a manifold the minimising geodesic to X bends, its foot slides, and the natural "margin" one would monotonically accumulate along the flow genuinely oscillates as the active contact set changes (Remark 3.4). This is the analytic heart of the difficulty and the reason a naive gradient-flow argument stalls exactly at the cut points, where it is asked to continue past a singularity of d_X. A further, quieter difficulty compounds it: M_X^g is *not closed* — minimising geodesics can merge in the limit at focal degenerations — so "the first time the flow meets the axis" need not exist.

The resolution, developed below, is to change the question the flow is asked to answer. Rather than flowing *through* the axis and tracking a monotone quantity, we flow *up to* the axis and read off a margin that is exact on the complement of the axis and merely needs to survive to the *first approach* — a formulation indifferent to whether the axis is closed and to everything that happens at focal and conjugate points. This severs the argument from every regularity subtlety at cut points, and it is what makes the theory unconditional.

1.4 Results

The paper proves three groups of theorems: metric, asymptotic, and topological.

Capture (§3). Theorem 3.1 shows that along any trajectory of the ascent inclusion $\Phi' \in -\operatorname{conv} U(\Phi)$, and up to the first approach to M_X^g, the distance to X grows at rate exactly one while the trajectory itself is 1-Lipschitz, so the starting point sits at depth $\geq d_X(p)$ inside every ball centred on the moving trajectory. The trajectory therefore either approaches M_X^g in finite time (*capture*: $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$) or escapes to infinity (*escape*, impossible on a compact manifold). Hence Corollary 3.2: $\mathcal{R}_X^g = M \setminus X$ on *every* compact manifold, for every closed $X \neq \varnothing$ — including a single point, which the two-point axis cannot detect at all. This answers the compact reconstruction question of [6, §5.2] affirmatively for the two-geodesic axis, with no hypothesis whatsoever. A spread refinement (Corollary 3.3) gives the two-point statement $\mathcal{R}_X = M \setminus X$ on every complete M with $\sup d_X < \operatorname{inj}(M)$.

Drift and the characterisation (§4). On a non-compact manifold, capture is not the whole story: a point may be reconstructed by remote balls whose centres run to infinity while its own trajectory escapes. Definition 2.4 packages the limits of these balls as *medial limit functions* —

limits $\psi = \lim[d(\cdot, a_k) - d_X(a_k)]$ of free-ball functions normalised by their own *reach* rather than by a basepoint — and Lemma 4.1 shows each reconstructs its whole open sublevel set, in three lines, on every complete manifold. Theorem 4.3 is the resulting characterisation,

$$\mathcal{R}_X^g = \bigcup \big\{ \{\psi < 0\} : \ \psi \text{ a medial limit function} \big\},$$

Theorem 4.4 the trichotomy *captured / drift-swallowed / escaping*, and Corollary 4.5 the induced reconstructibility criterion — the direct analogue, on an arbitrary complete manifold, of Białożyt's Theorem 1.1. On a Hadamard manifold the ideal medial limit functions are Busemann functions up to an additive level, and the reach normalisation absorbs what would otherwise be a delicate "radial offset" bookkeeping into the definition (Remark 4.7).

Specialisations (§5). We work out, in corollaries and worked examples, the flat picture (where the abstract characterisation aligns with Białożyt's theorem, its half-space limits identified by his defective-hyperplane certificate); total reconstruction on the sphere, on every spherical space form, and on every compact manifold, together with the two-point theorem on the round sphere (Corollary 5.4); the hyperbolic and Hadamard pictures (medial limit functions as Busemann functions at a level); and the non-compact positive-curvature examples where reconstruction is genuinely partial. Beyond these, we prove a faithful certificate on a manifold covered by neither Theorem 1.1 nor the Hadamard formulation: on the flat cylinder — flat but not simply connected — the free end-region above $\sup_X z$ is reconstructed in full if the top contact set omits a circle position, and not at all if it is the full circle (Proposition 5.11); a single contact point suffices, the wrap-around geodesic acting as the second contact.

The homotopy layer (§6). The escape class of the trichotomy turns out to be exactly the *Aubry set* of d_X in the sense of Cannarsa–Cheng–Fathi [10] (Lemma 6.4), so their theorem reads: the captured part of the complement always has the homotopy type of M_X^g (Theorem 6.5). When $\sup d_X < \infty$ — every compact M, every coarsely dense X — capture supplies the uniform clock that boundedness of the domain supplied for Lieutier [20] and Albano–Cannarsa–Nguyen–Sinestrari [2], and $M \setminus X \simeq M_X^g \simeq \mathrm{Cut}(X)$ on every complete manifold, components unbounded or not (Corollary 6.3). Two classical geometries fall out: uniform capture recovers the Ford–Voronoi spines of finite-volume cusped hyperbolic manifolds — dual to the Epstein–Penner decomposition [15] — and the Voronoi spines of crystallographic quotients (Corollary 6.7); and for a smooth closed submanifold X the captured part of the complement is, with no further hypothesis, the mapping cylinder of the swallow map onto the cut locus — a strong deformation retraction, with the Thom-space collapse of Basu–Prasad [4] as its inward dual (Proposition 6.8). The flat cylinder with $|X| = 1$ shows the bound $\sup d_X < \infty$ is not cosmetic: the reconstruction is complete, $\mathcal{R}_X^g = C \setminus X$, yet a figure-eight complement faces a contractible axis (Remark 6.6) — *homotopy equivalence follows capture, and swallowing does not transmit it.*

The medial complex (§7). The final group of results repairs that leak, unconditionally. The engine is a covering theorem (Theorem 7.3): *the finite layer does not leak* — every swallowed-uncaptured point lies in an *ideal* free horoball, at depth no smaller than its depth in any finite medial ball. The proof runs the capture clock along the transition between the captured set and the Aubry set: either the clocks of nearby captured points blow up, dragging medial centres to infinity with the margin intact, or the transition ray grazes the cut locus — and then rides it to infinity, with the same effect. Consequently $\mathcal{R}_X^g$ is the captured set plus the ideal cells (Corollary 7.4), and the *medial complex* — M_X^g with the ideal free horoballs glued along the diagram of the cover — always carries the homotopy type of $\mathcal{R}_X^g$, of $M \setminus X$ whenever the reconstruction is complete (Theorem 7.5): on the cylinder it rebuilds the figure-eight from a contractible axis and two ideal circles. The leak the complex repairs is not a curvature phenomenon: three points in the plane already reconstruct completely while their complement, a wedge of three circles, faces a contractible Voronoi tripod — the loops living in the corner cones where two ideal half-planes overlap inside the Aubry set (Proposition 7.8); for generic finite planar contact sets the leak has rank exactly the number of convex-hull corners (Remark 7.9).

Section 8 steps back and states plainly what the two layers of the theory — the *unconditional* dynamical layer, proved here in full, and the *faithful* certificate layer, of which Białożyt's theorem is the flat instance — do and do not deliver, and how the topological results sit across them. Section 9 collects what remains open, most of it on the faithful (escape) side.

1.5 Related work

The distance function from a closed set, its singular set, and its gradient dynamics have a substantial literature, and the detailed comparisons are placed where they bite: with the singular gradient-flow theorems of Lieutier [20] and Albano–Cannarsa–Nguyen–Sinestrari [2] and the propagation theory of Cannarsa–Cheng–Fathi [10] in Remark 3.5 and §6; with the critical point theory of distance functions of Grove–Shiohama [19, 18, 12] and the λ-medial axis of Chazal–Lieutier [11] in Remark 3.6; with the structure theory of the cut locus of a closed set of Tanaka–Sabau [26] and Miura–Tanaka [22] in §2; and with the cut-locus topology of submanifolds of Basu–Prasad [4, 24] in §6. What is new here, in one sentence each: the capture theorem extracts a quantitative reconstruction margin, for an arbitrary closed set in an arbitrary complete manifold, by *stopping* at the (non-closed) axis where the flow theorems are designed to continue; the medial limit functions make the escaping balls a theorem-free limit object on any complete manifold; and the homotopy results localise the failure of topological faithfulness in the Aubry set and repair it with the ideal cells of the medial complex.

Throughout, we illustrate the definitions and theorems with explicit examples in $\mathbb{R}^2$ and $\mathbb{R}^3$, on the round sphere, in the hyperbolic plane, on surfaces of revolution, and on the flat cylinder; the examples are chosen so that every phenomenon the general theory permits — capture, finite and ideal swallowing, escape, shielding, homotopy leak and its repair — is witnessed by a computation the reader can check by hand.

2 Preliminaries

Throughout, M is a complete Riemannian manifold; by the Hopf–Rinow theorem it is a proper geodesic metric space (closed balls are compact, and any two points are joined by a minimising geodesic). $X \subseteq M$ is closed with $\varnothing \neq X \neq M$, and $d_X = \mathrm{dist}(\cdot, X)$ is the distance to X, a 1-Lipschitz function, positive exactly off X. All geodesics are parametrised by arc length. We state the standing hypotheses once, to prevent a misreading the subject invites: *complete metric, closed X — nothing else.* No curvature assumption of any kind (sign, constancy, pinching, bounds) appears in §2–§4; the curvature of M may vary and change sign freely.

2.1 The two axes and the two reconstructions

For $x \in M \setminus X$ a *minimising geodesic to X* is a unit-speed geodesic $\gamma : [0, d_X(x)] \to M$ with $\gamma(0) = x$ and $\gamma(d_X(x)) \in X$; its *foot* is $\gamma(d_X(x))$. Set

$$U(x) = \{\gamma'(0) : \ \gamma \text{ a minimising geodesic to } X,\ \gamma(0) = x\} \subseteq T^1_x M,$$

the set of unit initial directions of such geodesics (a geodesic is determined by its initial vector, so distinct minimisers have distinct directions). Define the two medial axes

$$M_X = \{a : \ a \text{ has } \geq 2 \text{ nearest } \textit{points} \text{ in } X\}, \qquad M_X^g = \{x \in M \setminus X : \ |U(x)| \geq 2\},$$

the *two-point* and *two-geodesic* axes. Because two distinct minimising geodesics may share a foot, $M_X \subseteq M_X^g$; and M_X^g is the multi-geodesic part of the cut locus of X — in general a proper dense subset of it, the cut locus also containing focal points reached by a unique minimiser. Their reconstructions are

$$\mathcal{R}_X = \bigcup_{a \in M_X} B\big(a, d_X(a)\big), \qquad \mathcal{R}_X^g = \bigcup_{a \in M_X^g} B\big(a, d_X(a)\big), \qquad \mathcal{R}_X \subseteq \mathcal{R}_X^g,$$

unions of *open* free balls. When we use M_X^g as a target to be reached we write $\Sigma := M_X^g$; $r(a) = d_X(a)$ is the *reach* of a centre a. In $\mathbb{R}^n$ geodesics between points are unique, so $M_X = M_X^g$ and $\mathcal{R}_X = \mathcal{R}_X^g$, and the two theories coincide; the split is a purely curved phenomenon.

Remark 2.1 (Why the two-geodesic axis is the right object). The two-point axis M_X is the intuitive skeleton, but it is blind to a whole class of reconstructions. A single point $X = \{x_0\}$ on the round sphere has $M_X = \varnothing$ — every point off x_0 has the unique nearest point x_0 — yet the antipode $\bar{x}_0$ is a genuine medial centre: the entire sphere minus x_0 is a free ball about $\bar{x}_0$, because $\bar{x}_0$ is joined to x_0 by a sphere's worth of minimising geodesics, all of length π. Thus $\bar{x}_0 \in M_X^g \setminus M_X$ and $\mathcal{R}_X^g = \mathbb{S}^n \setminus \{x_0\}$ while $\mathcal{R}_X = \varnothing$. The reconstruction theory sees this only through M_X^g. This example, trivial as it is, already shows that the compact theorem below *cannot* hold for M_X (a single point is never reconstructed by two-point balls) and *must* be stated for M_X^g.

2.2 The direction field

Lemma 2.2 (Direction field: compactness and semicontinuity)**.** *For every $x \in M \setminus X$ the set $U(x)$ is nonempty and compact, and the multimap $x \mapsto U(x)$ is upper semicontinuous: if $x_k \to x$ in $M \setminus X$ and $u_k \in U(x_k)$, $u_k \to u$ (in TM), then $u \in U(x)$. In particular U is single-valued and continuous at every point of $(M \setminus X) \setminus \Sigma$.*

Proof. Nonemptiness and compactness: minimisers exist by properness, and a limit of initial vectors of minimising geodesics from the same x is the initial vector of a limit geodesic, which minimises by continuity of d_X and of lengths. Upper semicontinuity: let γ_k be the minimising geodesics with $\gamma_k'(0) = u_k$. By smooth dependence of geodesics on initial conditions, $\gamma_k \to \gamma$ locally uniformly where $\gamma'(0) = u$; the lengths $d_X(x_k) \to d_X(x)$, and the endpoints $\gamma_k(d_X(x_k)) \in X$ converge to $\gamma(d_X(x)) \in X$ (X closed). So γ is a geodesic from x to X of length $d_X(x)$: minimising, i.e. $u \in U(x)$. At a point with $U(x) = \{u\}$, upper semicontinuity of a compact-valued multimap into a singleton forces continuity of every selection near x — note this is continuity *at* x only; U may be multivalued arbitrarily close by. ☐

Lemma 2.3 (First variation of d_X; [21, §3])**.** *For $x \in M \setminus X$ and $w \in T_xM$, the one-sided directional derivative of d_X exists and equals*

$$\frac{d}{ds}\Big|_{s=0^+} d_X\big(\exp_x(sw)\big) = \min_{u \in U(x)} \langle w, -u \rangle \, .$$

Proof. "$\leq$": fix $u \in U(x)$ with geodesic γ, and fix a small $\delta > 0$. The point $\gamma(\delta)$ lies in a normal ball around x, so $q \mapsto d(q, \gamma(\delta))$ is smooth near x with gradient $-u$ at x. Since points of γ satisfy $d_X(\gamma(\delta)) = d_X(x) - \delta$,

$$d_X\big(\exp_x(sw)\big) \;\leq\; d\big(\exp_x(sw), \gamma(\delta)\big) + d_X(x) - \delta \;=\; d_X(x) + s\,\langle w,\, -u\rangle + O_\delta(s^2),$$

so $\limsup_{s\downarrow 0} s^{-1}[d_X(\exp_x(sw)) - d_X(x)] \leq \langle w,\, -u\rangle$ for every $u \in U(x)$, hence $\leq$ the minimum.

"$\geq$": let $s_k \downarrow 0$ realise the $\liminf$, put $x_k = \exp_x(s_k w)$, pick $u_k \in U(x_k)$ with geodesics γ_k, and set $\delta_k = \sqrt{s_k}$. The broken path through $\gamma_k(\delta_k)$ gives $d_X(x) \leq d(x, \gamma_k(\delta_k)) + d_X(x_k) - \delta_k$, and in normal coordinates at x_k, where $v_k := \exp_{x_k}^{-1}(x) = -s_k w + O(s_k^2)$,

$$d\big(x, \gamma_k(\delta_k)\big) \;\leq\; |\delta_k u_k - v_k| + O(\delta_k^3) \;\leq\; \delta_k - \langle v_k,\, u_k\rangle + \frac{|v_k|^2}{2\delta_k} + O(\delta_k^3) \;=\; \delta_k + s_k\,\langle w,\, u_k\rangle + O(s_k^{3/2}),$$

using $\sqrt{1+\tau} \leq 1 + \tau/2$ and the $O(r^2)$ metric distortion of normal coordinates. Hence $d_X(x_k) - d_X(x) \geq s_k\,\langle w,\, -u_k\rangle - O(s_k^{3/2})$. By Lemma 2.2 a subsequence of u_k converges to some $u \in U(x)$, so the $\liminf$ is $\geq \langle w,\, -u\rangle \geq \min_{u \in U(x)} \langle w,\, -u\rangle$. (Cf. [21, §3], where the semiconcavity of d_X on $M \setminus X$ — not needed here — is also established.) $\square$

Intuitively: d_X increases fastest, at unit rate, in the direction directly *away* from a nearest foot; at a two-geodesic point the rate in a given direction w is dictated by the *least favourable* competing foot, which is what makes the min appear and what makes the gradient flow turn.

Prior art on the distance from a closed set. The dichotomy beneath everything here — d_X is differentiable exactly where the foot direction is unique, singular exactly on Σ — is classical, and its fine structure theory is active. Tanaka and Sabau [26] characterise the differentiable points of the distance from an arbitrary closed subset in Finsler generality and prove that on surfaces its cut locus is a local tree, statements new even in the Riemannian case; Miura and Tanaka [22] prove a delta-convex structure theorem for the singular set. The capture argument needs none of this finer structure — only Lemmas 2.2 and 2.3 above — but the structure theory is what a regularity upgrade of the homotopy layer of §6, and a CW model of the medial complex of §7 (Remark 7.7(c)), would draw on.

2.3 The ascent inclusion and capture curves

On the open set $M \setminus X$ define the multimap

$$G(x) \;:=\; \operatorname{conv}\big(-U(x)\big) \;\subseteq\; T_x M,$$

the convex hull of the reversed minimising directions. By Lemma 2.2, G is upper semicontinuous with nonempty compact convex values contained in the closed unit ball. A *capture curve* from $p \in M \setminus X$ is a locally Lipschitz curve $\Phi : [0, T_{\max}) \to M \setminus X$ with $\Phi_0 = p$ and

$$\Phi'(t) \in G(\Phi_t) \qquad \text{for a.e. } t, \tag{1}$$

maximally extended in $M \setminus X$. Local existence of solutions of (1), and extension as long as the trajectory stays in a compact subset of the open domain, are the standard theory of differential inclusions with upper semicontinuous convex compact right-hand side (Aubin–Cellina [3, §2.1]), applied in charts and concatenated; $|\Phi'| \leq 1$ a.e., so trajectories are 1-Lipschitz. We emphasise that *no* uniqueness, canonicity, or semiconcave-gradient selection is needed: any solution serves. (At a point of Σ the inclusion may branch; off Σ it is the classical ODE of the field $-u$.) When U is single-valued along a trajectory the curve simply extends the current minimising geodesic beyond x, switching fields continuously as the foot slides.

2.4 Free-ball functions and medial limit functions

For $a \in M \setminus X$ set

$$\psi_a := d(\cdot, a) - d_X(a) \;:\; M \to \mathbb{R},$$

a 1-Lipschitz function, negative exactly on the open free ball $B(a, d_X(a))$, with $\psi_a \geq 0$ on X. It is normalised by the reach — $\min \psi_a = -d_X(a)$, attained at a — rather than by a basepoint; this intrinsic normalisation, "measure the ball by how deep it goes, not by how far its centre is", is what makes the limits below basepoint-free, and it is the technical device that dissolves the offset bookkeeping of the Hadamard theory (Remark 4.7).

Definition 2.4 (Medial limit functions)**.** *A function $\psi : M \to \mathbb{R}$ is a* medial limit function *of X if there are $a_k \in M_X^g$ with $\psi_{a_k} \to \psi$ locally uniformly. It is* finite *if (a_k) has a bounded subsequence and* ideal *otherwise (then ψ is a horofunction-type limit: $d(a_k, \cdot) \to \infty$ locally uniformly while ψ stays finite). Every ψ is 1-Lipschitz, and $\psi \geq 0$ on X (each ψ_{a_k} is), so $\{\psi < 0\}$ is an open set disjoint from X: the* free horoball *of ψ. For finite ψ along $a_k \to a$: if $a \in M_X^g$ then $\psi = \psi_a$ is a free-ball function; in general (M_X^g is not closed) $\psi = \psi_a$ with a a focal-degenerate limit of medial centres, or even a point of X itself (medial centres collapsing into a corner of X with $d_X(a_k) \to 0$), where $\psi_a = d(\cdot, a)$ and the free horoball is empty — still usable, by Lemma 4.1, which never needs the limit centre itself.*

Remark 2.5 (Relation to the horofunction compactification)*.* M proper makes $C(M)$ with the topology of locally uniform convergence a compact home for 1-Lipschitz functions normalised at a point (Arzelà–Ascoli). The classical horofunction compactification [17, 9] normalises $d(\cdot, a_k)$ by a basepoint value; our ψ's differ from it by the scalar $d(o, a_k) - d_X(a_k)$ — the *offset*. Working with reach-normalised limits absorbs the offset into the function itself; see Remark 4.7. On a Hadamard manifold every ideal medial limit function is a Busemann function plus a constant (horofunctions of a proper CAT(0) space are Busemann [9, II.8.13]), so ideal medial limit functions are exactly "Busemann function b_ξ + level c".

3 The capture theorem

The capture theorem is the engine of the compact case and one of the two inputs to the general characterisation. Its statement is a clean dichotomy; its proof is a one-line differential identity valid off the axis, plus a compactness argument for the escape alternative.

Theorem 3.1 (Capture or escape)**.** *Let M be complete, X closed, $p \in M \setminus X$, and let Φ be any capture curve from p; assume $p \notin \Sigma$ (otherwise p is captured at time 0: $a = p$ contains p at depth $d_X(p)$ in its own medial ball). Let*

$$T \;:=\; \sup\bigl\{t \in [0, T_{\max}) : \; \Phi_s \notin \Sigma \quad \forall s \in [0, t]\bigr\} \;\in\; [0, T_{\max}]$$

($T = 0$ is possible: Σ need not be closed, and the trajectory may approach it instantly; this lands in case (i)*). Then on $[0, T)$,*

$$d_X(\Phi_t) = d_X(p) + t, \qquad d(p, \Phi_t) \leq t, \qquad \textit{hence} \qquad d_X(\Phi_t) - d(p, \Phi_t) \;\geq\; d_X(p), \tag{2}$$

and exactly one of the following holds.

(i) (Capture.) $T < T_{\max}$. *Then there exist* $a \in M^g_X$ *arbitrarily close to* Φ_T *(*$a = \Phi_T$ *itself if* $\Phi_T \in \Sigma$*) with*

$$d_X(a) - d(p,a) \ \geq \ d_X(p) - \varepsilon \qquad (\varepsilon > 0 \textit{ arbitrary}),$$

so $p \in B(a, d_X(a))$ *for such* a*:* $\quad p \in \mathcal{R}^g_X$*, at depth arbitrarily close to* $d_X(p)$*, and* $T \leq \operatorname{diam}(M) - d_X(p)$*.*

(ii) (Escape.) $T = T_{\max}$. *Then* $T_{\max} = \infty$*, the trajectory never meets* Σ*,* $d_X(\Phi_t) = d_X(p) + t \to \infty$*, and* Φ_t *leaves every compact subset of* M*. This is possible only if* M *is non-compact.*

Proof. The rate-one identity. Fix $t \in [0, T)$; on $[0,t]$ the trajectory avoids Σ, so $U(\Phi_s) = \{u(\Phi_s)\}$ is a singleton for every $s \leq t$ and (1) reads $\Phi'(s) = -u(\Phi_s)$ at a.e. s. The map $h(s) := d_X(\Phi_s)$ is Lipschitz. Let s be any point where both $\Phi'(s)$ exists (with the inclusion valid) and $h'(s)$ exists — almost every $s \in [0,t]$. Write $v = \Phi'(s) = -u(\Phi_s)$, a unit vector. In a chart around Φ_s, differentiability gives $d\big(\Phi_{s+\sigma}, \exp_{\Phi_s}(\sigma v)\big) = o(\sigma)$ as $\sigma \downarrow 0$, and d_X is 1-Lipschitz, so

$$h(s+\sigma) = d_X\big(\exp_{\Phi_s}(\sigma v)\big) + o(\sigma) = h(s) + \sigma \min_{u \in U(\Phi_s)} \langle v, -u \rangle + o(\sigma) = h(s) + \sigma \ \langle -u, -u \rangle + o(\sigma),$$

by Lemma 2.3 and single-valuedness. Hence $h'(s) = |u|^2 = 1$ at a.e. $s \in [0,t]$, and integrating, $h(t) = d_X(p) + t$. The bound $d(p, \Phi_t) \leq t$ is the 1-Lipschitz property of the trajectory, and the margin inequality in (2) is their difference. Note that no closedness of Σ, and no structure at focal or conjugate points, was used: a point of $(M \setminus X) \setminus \Sigma$ in the closure of Σ is simply a point where U is single-valued, and the identity passes through it.

Dichotomy. Suppose first $T < T_{\max}$. If $\Phi_T \in \Sigma$, continuity of $d_X(\Phi.) - d(p, \Phi.)$ and (2) on $[0,T)$ give $d_X(\Phi_T) - d(p, \Phi_T) \geq d_X(p) > 0$, so $a = \Phi_T \in M^g_X$ contains p in its open medial ball at depth $\geq d_X(p)$. If $\Phi_T \notin \Sigma$, then by definition of T as a supremum, every interval $(T, T+\delta] \subseteq [0, T_{\max})$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$; letting $\delta \downarrow 0$ and using continuity of the margin, $a = \Phi_{s_\delta} \in M^g_X$ satisfies $d_X(a) - d(p,a) \geq d_X(p) - \varepsilon$ for δ small. In both cases $p \in B(a, d_X(a))$ with $a \in M^g_X$, i.e. $p \in \mathcal{R}^g_X$; and $d_X \leq \operatorname{diam}(M)$ bounds T via (2).

Now suppose $T = T_{\max}$: the trajectory never meets Σ, and (2) holds on all of $[0, T_{\max})$. If $T_{\max} < \infty$, then $\Phi([0, T_{\max})) \subseteq \bar{B}(p, T_{\max}) \cap \{d_X \geq d_X(p)\}$, a compact subset of the open set $M \setminus X$ (properness of M); by the extension property of solutions this contradicts maximality. So $T_{\max} = \infty$ and $d_X(\Phi_t) = d_X(p) + t \to \infty$; since d_X is bounded on compact sets, Φ_t leaves every compact set. On compact M, $d_X \leq \operatorname{diam}(M)$ makes this impossible, so case (i) occurs. $\square$

Corollary 3.2 (Total reconstruction on compact manifolds, unconditional)**.** *Let* M *be a compact Riemannian manifold and* $X \subseteq M$ *closed,* $\varnothing \neq X \neq M$*. Then*

$$\boxed{\mathcal{R}^g_X = M \setminus X},$$

and quantitatively: every $p \in M \setminus X$ *lies at depth* $\geq d_X(p) - \varepsilon$ *(any* $\varepsilon > 0$*) inside a medial ball of* M^g_X*. No curvature, genericity, analyticity, spread, or cardinality hypothesis is required; in particular* $|X| = 1$ *is allowed, where the two-point axis has* $M_X = \varnothing$ *but the two-geodesic axis still reconstructs everything.*

Proof. $\mathcal{R}^g_X \subseteq M \setminus X$ since medial balls are free. Conversely, if $p \in M^g_X$ then $p \in B(p, d_X(p)) \subseteq \mathcal{R}^g_X$; otherwise start a capture curve at p (local existence, §2); by Theorem 3.1 escape is impossible on compact M, so p is captured. $\square$

In particular the reconstruction question on compact Riemannian manifolds, raised for the medial axis in [6, §5.2], has an unconditional affirmative answer for the two-geodesic axis.

Corollary 3.3 (Two-point axis under spread)**.** *Let M be complete with* $\mathrm{inj}(M) := \inf_{q\in M} \mathrm{inj}(q) > 0$*, and suppose*

$$\sup_{M\setminus X} d_X \;<\; \mathrm{inj}(M) \qquad \text{— e.g. } X \ \delta\text{-dense with } \delta < \mathrm{inj}(M).$$

Then $\mathcal{R}_X = M \setminus X$ for the two-point axis. Compactness is not needed: it would enter only to preclude escape, which the spread hypothesis already does. In $\mathbb{R}^n$ ($\mathrm{inj} = \infty$*) the statement recovers the classical fact that every coarsely dense closed set reconstructs its complement:* $\sup d_X < \infty$ *implies $\mathcal{R}_X = \mathbb{R}^n \setminus X$.*

Proof. Escape is impossible: along an escaping capture curve, $d_X(\Phi_t) = d_X(p) + t \to \infty$ would exceed $\sup d_X$. So every $p \in (M \setminus X) \setminus M_X^g$ is captured (Theorem 3.1(i), within time $\sup d_X - d_X(p)$), giving $a \in M_X^g$ with $p \in B(a, d_X(a))$; for $p \in M_X^g$ take $a = p$. Two distinct minimising geodesics from a with the same endpoint x_0 would give two distinct vectors of norm $d(a, x_0) = d_X(a) < \mathrm{inj}(M) \le \mathrm{inj}(a)$ with the same image under $\exp_a$, contradicting injectivity of $\exp_a$ on the open ball of radius $\mathrm{inj}(a)$. So the two geodesics of a have distinct endpoints: $a \in M_X$. ☐

Remark 3.4 (Why stopping at Σ, not flowing through it, is the correct formulation)*.* Two natural strengthenings of Theorem 3.1 are false, and identifying their failure is what shaped the proof. First, *the margin $d_X(\Phi) - d(p,\Phi)$ is not monotone through multi-geodesic arcs*, already in $\mathbb{R}^2$: for $X = \{(0,0),(0,2)\}$ and $p = (1, 0.9)$, the Lieutier flow reaches the bisector $\{y = 1\}$ and slides along it, and the margin decreases strictly (from ≈ 1.345 toward the limit 1, the depth of p in the limiting half-plane). Second, even the quadratic margin $d_X^2 - d(p,\cdot)^2$, which *is* monotone along unique-foot and bisector arcs in $\mathbb{R}^n$, fails at turning points where the active contact set changes (a three-point configuration in $\mathbb{R}^2$ suffices: $X = \{(0,0),(0,2),(10,5)\}$, with the flow from $p = (1, 0.9)$ turning at $\approx (5.75, 1)$, where the derivative of the quadratic margin is negative). Both phenomena live *on* Σ; on its complement the rate-one identity (2) is exact. Since reaching Σ is the goal, not an obstacle, the correct move is to stop there — capture is achieved, with the margin intact, *before* the first multi-geodesic time. The failure of closedness of Σ (focal degenerations, e.g. the evolute of an ellipse in $\mathbb{R}^2$, or the conjugate-cusp endpoints of an ellipsoid's cut segment) is equally harmless: such points are unique-geodesic points, the field $-u$ is continuous at them (Lemma 2.2), and the trajectory passes through with d_X still growing at rate one, the foot sliding continuously.

Remark 3.5 (Relation to the singular gradient flows)*.* For semiconcave d_X one can canonically select the minimal-norm element of G (Lieutier's generalised gradient [20], in $\mathbb{R}^n$; [1, 21] for the semiconcavity). Two theorems run that canonical flow *into* the singular set: Lieutier's homotopy equivalence of a bounded domain with its medial axis [20], and its Riemannian counterpart by Albano, Cannarsa, Nguyen and Sinestrari [2], who prove that the singular set of the boundary distance of a bounded domain on a Riemannian manifold is invariant under the generalised gradient flow and deduce the homotopy equivalence of domain and singular set; see Cannarsa–Cheng–Fathi [10] for the current form of the propagation-of-singularities method behind such results. The rate-one growth of d_X off the singular set is classical in that literature; what Theorem 3.1 takes from the flow is different, and in a sense opposite. Our argument analyses the inclusion only *off* Σ, where every selection agrees with the naive one (extend the unique minimising geodesic); it imposes no semiconcave or minimal selection — any relaxed solution serves; and it *terminates* at Σ, where those flows are designed to continue. The deliverable is not a retraction but the quantitative margin $d_X(a) - d(p,a) \geq d_X(p)$ at the first approach — a reconstruction statement with no counterpart in the homotopy results — extracted on an arbitrary complete manifold, for an arbitrary closed X, with the escape alternative live. This is why no analogue of the swallow property past cut points is needed: the trajectory is stopped at first approach, and the reconstruction is read off the margin accumulated strictly before it. The price — capture produces a centre only *near* the first Σ-approach when $\Phi_T \notin \Sigma$ — costs an arbitrarily small ε of depth, invisible to the open-ball statement. Section 6 completes the comparison in the opposite direction: importing their invariance theorem, capture supplies the uniform clock that boundedness of the domain supplied in [20, 2], extending the homotopy equivalence to every complete M with $\sup d_X < \infty$; and the Aubry set of [10] turns out to be exactly the escape class of Theorem 4.4, which locates the failure beyond that bound.

Remark 3.6 (Capture as critical point theory of d_X). The inclusion (1) is also the engine of the critical point theory of distance functions of Grove–Shiohama [19, 12, 18]: $q \in M \setminus X$ is *critical* for d_X iff $0 \in \operatorname{conv} U(q)$, which forces $|U(q)| \geq 2$ — so all critical points of d_X lie on M_X^g — and the isotopy lemma deforms sublevel sets across critical-value-free intervals along precisely such gradient-like curves. In that vocabulary Theorem 3.1 is an isotopy lemma made quantitative and unconditional: it assumes no critical-value-free interval — the trajectory is stopped at its first approach to the (non-closed) set $M_X^g \supseteq \{\text{critical points}\}$, whenever that occurs — and it returns not an isotopy but the margin $d_X(a) - d(p,a) \geq d_X(p)$: a reconstruction statement. The min-type Morse theory of Gershkovich–Rubinstein [16] structures the same objects on nonpositively curved manifolds, where germs of d_X are min-type; and in $\mathbb{R}^n$ the *weak feature size* of Chazal–Lieutier [11] — the smallest critical value of d_X, below which their λ-medial axis carries the homotopy type of the complement — is the global critical scale whose pointwise, per-trajectory analogue is the capture margin. A Riemannian λ-medial axis built on that margin is a natural project we do not open here.

4 Drift: the swallow, the characterisation, the trichotomy

Capture is complete on compact manifolds, but on a non-compact M a point can be reconstructed by medial balls whose centres go to infinity while its own capture curve escapes. The primal example is $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$: a point p in an open half-plane bounded by the line x_0x_1, placed beyond x_0 at an obtuse angle, has a capture curve that runs straight to infinity with the single foot x_0 — yet p is swallowed by remote bisector balls $B(a_k, r(a_k))$: a single sufficiently distant ball already contains it, and so does every later member of the escaping sequence $a_k \to \infty$. The limit of those balls is a half-plane, and the correct general home for such limits is the space of medial limit functions of Definition 2.4.

Lemma 4.1 (Swallow, general and offset-free)**.** *Let M be any complete Riemannian manifold, X closed, and ψ a medial limit function of X. Then*

$$\{\psi < 0\} \;\subseteq\; \mathcal{R}_X^g.$$

If the witnessing centres lie in M_X, then $\{\psi < 0\} \subseteq \mathcal{R}_X$.

Proof. Let $\psi = \lim \psi_{a_k}$ locally uniformly, $a_k \in M_X^g$, and let $q \in \{\psi < 0\}$. Then $\psi_{a_k}(q) \to \psi(q) < 0$, so for large k

$$d(q, a_k) - d_X(a_k) = \psi_{a_k}(q) < 0,$$

i.e. $q \in B(a_k, d_X(a_k))$ with $a_k \in M_X^g$: $q \in \mathcal{R}_X^g$. $\square$

There is no anchoring of feet, no curvature bound, no pinching, no offset hypothesis: the entire content of the swallowing step — which in curved settings historically required delicate control of where the escaping centres' feet accumulate — is absorbed into the normalisation of ψ_a by the reach.

Lemma 4.2 (Compactness of reconstructing sequences)**.** *Let $p \in \mathcal{R}_X^g$, and let $a_k \in M_X^g$ be any sequence with $\limsup_k\big[d_X(a_k) - d(p, a_k)\big] > 0$ (e.g. a sequence of centres whose balls contain p with margin bounded away from 0; a single ball repeated qualifies). Then a subsequence of (ψ_{a_k}) converges locally uniformly to a medial limit function ψ with $\psi(p) < 0$.*

Proof. The functions ψ_{a_k} are 1-Lipschitz and $\psi_{a_k}(p) = d(p, a_k) - d_X(a_k)$ is bounded: below by $-d_X(p)$ (since $d_X(a_k) \le d_X(p) + d(p, a_k)$) and above by 0 along the qualifying subsequence. By Arzelà–Ascoli on the proper space M, a further subsequence converges locally uniformly; the limit is a medial limit function, and $\psi(p) = \lim \psi_{a_k}(p) \le -\limsup[d_X(a_k) - d(p, a_k)] < 0$. $\square$

Theorem 4.3 (Characterisation on an arbitrary complete manifold)**.** *For every complete Riemannian manifold M and every closed $X \subseteq M$, $\varnothing \ne X \ne M$,*

$$\boxed{\mathcal{R}_X^g \;=\; \bigcup\big\{\{\psi < 0\} : \;\psi \text{ a medial limit function of } X\big\}}.$$

The same identity holds for $\mathcal{R}_X$ and M_X in place of $\mathcal{R}_X^g$ and M_X^g. (Every finite medial limit function is itself of the form $\psi_a = d(\cdot, a) - d_X(a)$ with a in the closure of M_X^g taken in M: a focal-degenerate limit of medial centres, or — degenerately — a point of X, where $d_X(a) = 0$ and the sublevel set is empty. Either way its sublevel set is swallowed by the lemma, which never needs the limit centre itself.)

Proof. "$\supseteq$" is Lemma 4.1. "$\subseteq$": if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$, and ψ_a — a medial limit function via the constant sequence — has $\psi_a(p) < 0$. $\square$

The identity is deliberately economical, and one should read its force correctly. Its finite members alone give a tautology — every ball is its own limit — so the set equality is not the point. The point is the pair of facts it packages. *First*, the *ideal* members reconstruct their entire open sublevel set, bare, with no anchoring or pinching hypothesis: this is Lemma 4.1 applied to horofunction-type limits. *Second*, combined with Theorem 3.1 it yields a clean trichotomy.

Theorem 4.4 (Trichotomy on a complete manifold)**.** *Let M be complete, X closed, $p \in M \setminus X$. Exactly one of the following holds.*

1. (Captured.) *Some capture curve from p approaches M_X^g in finite time. Then $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$.*
2. (Swallowed, uncaptured.) *No capture curve from p approaches M_X^g in finite time, but $\psi(p) < 0$ for some medial limit function ψ. Then $p \in \mathcal{R}_X^g$: reconstructed, but only by centres p's own ascent cannot find — a remote ball, or a drifting sequence.*
3. (Escaping.) *Neither. Then $p \notin \mathcal{R}_X^g$, and every capture curve from p escapes to infinity with $d_X(\Phi_t) = d_X(p) + t$.*

In particular $\mathcal{R}_X^g = \{p :$ (1) or (2)$\}$, and on compact M class (1) *is all of $M \setminus X$.*

Proof. The classes are exclusive and exhaustive by construction. (1) $\Rightarrow$ $p \in \mathcal{R}^g_X$ is Theorem 3.1(i); (2) $\Rightarrow p \in \mathcal{R}^g_X$ is Lemma 4.1. Conversely, if $p \in \mathcal{R}^g_X$ then $p \in B(a, d_X(a))$ for some $a \in M^g_X$ and the finite medial limit function ψ_a has $\psi_a(p) < 0$, so p lies in class (2) whenever it is not in class (1) — never in class (3). Finally, a point not in class (1) has every capture curve uncaptured, hence escaping, by the dichotomy of Theorem 3.1. $\square$

Białożyt's theorem is not a description of the reconstructed set but a *reconstructibility criterion*: it says exactly when a closed set is recovered in full. The trichotomy yields the direct analogue on any complete manifold, because $\mathcal{R}^g_X$ is precisely classes (1)–(2) while class (3) is its complement.

Corollary 4.5 (Reconstructibility criterion)**.** *Let M be complete, X closed, $\varnothing \neq X \neq M$. The following are equivalent.*

(a) *X is reconstructible: $\mathcal{R}^g_X = M \setminus X$.*

(b) *$M \setminus X \;\subseteq\; \bigcup\bigl\{\{\psi < 0\} : \;\psi$ a medial limit function of $X\bigr\}$.*

(c) *Every $p \in M \setminus X$ is captured or drift-swallowed; equivalently, the escaping class* (3) *of Theorem 4.4 is empty.*

(d) *For every $p \in M \setminus X$ whose capture curves escape to infinity, some medial limit function ψ satisfies $\psi(p) < 0$.*

Proof. $(a) \Leftrightarrow (b)$: by Theorem 4.3 the right-hand side of (b) equals $\mathcal{R}^g_X$, and $\mathcal{R}^g_X \subseteq M \setminus X$ always, so (b) is the reverse inclusion, i.e. (a). $(a) \Leftrightarrow (c)$: by Theorem 4.4, $\mathcal{R}^g_X$ is exactly classes (1)–(2) and is disjoint from class (3); since the three classes partition $M \setminus X$, $\mathcal{R}^g_X = M \setminus X$ iff class (3) is empty. $(c) \Leftrightarrow (d)$: a point outside class (1) escapes (Theorem 3.1), so it lies in class (2) — swallowed — iff some ψ has $\psi(p) < 0$; captured points lie in $\mathcal{R}^g_X$ regardless, so emptiness of class (3) is exactly (d). $\square$

Two remarks fix its meaning. First, *on a compact manifold the criterion is vacuously satisfied*: escape is impossible (Theorem 3.1), class (3) is empty, and X is reconstructible for free (Corollary 3.2). The criterion therefore has content only on non-compact M — exactly where Białożyt's does, $\mathbb{R}^n$ being non-compact. Second, condition (b) is the *unconditional* form: it refers to the medial limit functions, hence implicitly to the axis M^g_X. Białożyt's theorem is the *faithful* sharpening of (b) in the flat case, replacing "$M \setminus X \subseteq \bigcup\{\psi < 0\}$" by the axis-free hull condition

$$\mathbb{R}^n \setminus X \;\subseteq\; \overline{\operatorname{conv}}\, X \;\cup \bigcup_{L \text{ defective}} L^+$$

of Theorem 1.1 (Corollary 5.1); on a Hadamard manifold the corresponding faithful sharpening replaces the ideal sublevel sets by defective horoballs $\{b_\xi < -c(\xi)\}$ (§5.3, Remark 4.7). Thus Corollary 4.5 is the general analogue of Białożyt's main theorem, and the passage from (b) to a checkable-from-X certificate is precisely the faithful-layer problem: solved in flat geometry, formulated in Hadamard geometry (§5.3), and open in mixed curvature (§9).

Remark 4.6 (The witnesses of class (2)). Both kinds of witness genuinely occur, already for $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$ and p beyond x_0 in an open half-plane, at an obtuse angle to the bisector: every capture curve from p runs straight to infinity with the single foot x_0, yet p lies in a single *finite* bisector ball (and also in the ideal half-plane limit of the bisector balls). Class (2) is thus "reconstruction is non-local for p": the certificate cannot be found by ascending from p, which is exactly the phenomenon that separates the non-compact theory from the compact one and forces the faithful certificates below to quantify over supporting half-spaces or horoballs rather than over flow lines.

Remark 4.7 (The Hadamard offset, dissolved). Let M be Hadamard (complete, simply connected, sec ≤ 0). Ideal medial limit functions are exactly $b_\xi + c$ with $\xi \in \partial_\infty M$ and $c \in \mathbb{R}$ (Remark 2.5), normalised so that b_ξ vanishes on the horosphere of the maximal free horoball $\mathcal{H}$ at ξ; necessarily $c \geq 0$ when X meets every neighbourhood of that horosphere. Call $\mathcal{H}$ *defective at level* c if $b_\xi + c$ is a medial limit function; then Lemma 4.1 reconstructs $\{b_\xi < -c\}$ — the horoball shrunk by c. Relative to a basepoint o, the same drift carries a scalar "radial offset" $\mathrm{off}_k = d(o, a_k) - d_X(a_k)$ — the curved form of the β of §5.1 — through the exact identity $\psi_{a_k} = \big[d(\cdot, a_k) - d(o, a_k)\big] + \mathrm{off}_k$: the bracket converges to the basepoint-normalised horofunction, so the level of the limit is the limit offset, re-based to the horosphere of $\mathcal{H}$. What in a basepoint-normalised treatment looks like a gap — "only the deep part $\{b_\xi < -c\}$ is reconstructed when the asymptotic offset does not vanish" — is simply the content of the level: the drift *is* a level-c certificate and reconstructs exactly $\{b_\xi < -c\}$, while the full-horoball statement is the assertion that a level-0 certificate exists. The exact level

$$c(\xi) \;:=\; \inf\{c \geq 0 : \; b_\xi + c \text{ is a medial limit function}\}$$

is a question about X and the horosphere geometry, not about the medial axis. The level vanishes in every instance computed in this paper — the flat case (Corollary 5.1) and the cylinder ends (Remark 5.12) — and no example with $c(\xi) > 0$ is known (§9(2)).

5 Specialisations and worked examples

We now specialise Theorems 4.3–4.4 to the classical geometries and illustrate every key definition with an explicit example. Throughout, the *unconditional* content — the abstract characterisation — is supplied by the theorems above; the *faithful* content — a certificate readable from X and M — is supplied by Białożyt's theorem in the flat case, is vacuous in the compact case, and is proved here for the ends of the flat cylinder (§5.5), the first such certificate beyond Euclidean space.

5.1 Euclidean space: Białożyt's theorem as the flat instance

In $\mathbb{R}^n$ geodesics between points are unique, so $M_X^g = M_X$ and $\mathcal{R}_X^g = \mathcal{R}_X$. We identify the medial limit functions. A finite medial limit function is a free-ball function $y \mapsto |y - a| - r$ with a a limit of medial centres. For an *ideal* one, let $a_k \to \infty$ along a unit direction u (i.e. $a_k/|a_k| \to u$) with reaches $r_k = d_X(a_k)$. Then, for each fixed y,

$$\psi_{a_k}(y) = |y - a_k| - r_k = \big(|y - a_k| - |a_k|\big) + \big(|a_k| - r_k\big) \longrightarrow \; -\langle y, \, u\rangle + \beta,$$

where the first bracket tends to $-\langle y, \, u\rangle$ (a standard Euclidean computation) and convergence of ψ_{a_k} forces the offset $|a_k| - r_k$ to a finite limit β. So every ideal medial limit function is *affine*,

$$\psi(y) = \beta - \langle y, \, u\rangle \,, \qquad \{\psi < 0\} = \{\langle y, \, u\rangle > \beta\},$$

an open half-space with outward normal u. Conversely a supporting half-space limit of medial balls is exactly such a ψ. Theorem 4.3 therefore reads, in $\mathbb{R}^n$:

$$\mathcal{R}_X \;=\; \Big(\bigcup \text{finite medial balls}\Big) \;\cup\; \Big(\bigcup\{\text{open half-spaces that are ideal medial limits}\}\Big).$$

This is Białożyt's decomposition. The finite balls fill $\overline{\mathrm{conv}}\, X \setminus X$ — every point of the closed hull off X lies in a free medial ball [6]; the half-spaces are the outer half-spaces L^+ of supporting hyperplanes L that arise as ideal limits. The analysis of [6] identifies *which* supporting hyperplanes occur: precisely the *defective* ones, $X \cap L \neq \overline{\mathrm{conv}}\, X \cap L$ (equivalently, by Motzkin's theorem [23], those carrying a non-Chebyshev point just outside the contact face). Thus:

Corollary 5.1 (Flat specialisation)**.** *In $\mathbb{R}^n$, Theorem 4.3 together with the two Euclidean inputs of [6] just quoted — the closed hull off X is covered by free medial balls, and the ideal half-space limits are exactly the outer half-spaces of the defective supporting hyperplanes — gives*

$$\mathcal{R}_X \;=\; \big(\overline{\mathrm{conv}}\, X \setminus X\big) \;\cup \bigcup_{L \ \textit{defective}} L^+,$$

and the reconstructibility criterion this yields — X is reconstructible iff $\mathbb{R}^n \setminus X \subseteq \overline{\mathrm{conv}}\, X \cup \bigcup_L L^+$ — is exactly Theorem 1.1. The drift over a defective L runs at level 0 relative to L (β equals the support value of L): the escaping bisector centres accumulate their feet on the defective contact face, and their half-space limit is L^+ itself, unshrunk. The escaping class (3) of Theorem 4.4 is exactly Białożyt's unreconstructed set — the rays leaving X with a single foot and no defective facet around them.

Here the split of labour is exact and instructive: the abstract characterisation (unconditional) delivers "$\mathcal{R}_X$ is balls $\cup$ half-space limits"; Białożyt's certificate (the faithful layer) delivers "the half-space limits are exactly the defective supporting hyperplanes." This is the template the whole paper follows on other geometries.

Example 5.2 (Two points)*.* $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$, so $\overline{\mathrm{conv}}\, X$ is the segment $[x_0, x_1]$ and the axis M_X is its perpendicular bisector. There is exactly one defective supporting line: the line ℓ *through* x_0x_1 itself, on which $\overline{\mathrm{conv}}\, X \cap \ell = [x_0, x_1]$ while $X \cap \ell = \{x_0, x_1\}$ — two points, not the whole segment. It supports from both sides, so it is defective in both orientations, with the two open half-planes bounded by ℓ as outer half-spaces L^+; these are the ideal (half-plane) medial limits of the escaping bisector balls. By Corollary 5.1,

$$\mathcal{R}_X = \big([x_0, x_1] \setminus \{x_0, x_1\}\big) \;\cup\; \{\text{two open half-planes}\} \;=\; \mathbb{R}^2 \setminus \big(\ell \setminus (x_0, x_1)\big),$$

i.e. *everything except the two outer rays* of ℓ beyond the endpoints (together with X itself). Note what is and is not reconstructed: the *open* segment (x_0, x_1) *is* reconstructed — by finite bisector balls, which contain every interior segment point — while the two collinear rays $\{x_0 + t(x_0 - x_1) : t > 0\}$ and its mirror are on *no* free ball and are missed by the open half-planes. The failure of reconstructibility for a two-point set is exactly the failure to cover these rays. This is the primal drift example of Remark 4.6: a point in an open half-plane at an obtuse angle to the bisector is reconstructed only through the bisector balls and their ideal half-plane limit (class (2)), not by its own escaping capture curve.

Example 5.3 (A triangle: the outer half-spaces close up)*.* Let X be the three vertices of an equilateral triangle, so $\overline{\mathrm{conv}}\, X$ is the solid triangle. A supporting line through a single vertex has $X \cap L = \overline{\operatorname{conv}}\, X \cap L = \{v\}$ — not defective. A supporting line *along an edge* has $\overline{\operatorname{conv}}\, X \cap L$ the whole edge while $X \cap L$ is its two endpoints — defective, with a genuinely nonempty outer half-plane L^+ reconstructed by the ideal drift of the escaping bisector centres of those two vertices (which are legitimate medial centres far out beyond the edge, the third vertex being strictly farther). The three edge half-planes *overlap to cover every exterior point* — a point beyond a vertex lies in the outer half-plane of an incident edge — and the interior is filled by finite balls, so $\mathcal{R}_X = \mathbb{R}^2 \setminus X$: reconstructible. Contrast Example 5.2, the degenerate "triangle" with one edge: there the single edge's two outer half-planes leave the collinear rays uncovered, and reconstructibility fails. The moral is that reconstructibility is about whether the defective outer half-spaces *tile* the exterior, which a genuine polygon does and a segment does not. (The same three points return in Proposition 7.8, where complete reconstruction turns out *not* to imply that the axis carries the homotopy type of the complement.)

5.2 Compact manifolds, the sphere, and space forms

Here the faithful layer is *vacuous*: on a compact manifold M^g_X is bounded, no sequence of medial centres escapes, there are no ideal medial limit functions and no hull condition, and Corollary 3.2 gives total reconstruction outright.

Corollary 5.4 (Round sphere and spherical space forms)**.** (a) *For every closed* $\varnothing \neq X \neq \mathbb{S}^n$*,* $\mathcal{R}^g_X = \mathbb{S}^n \setminus X$*; likewise* $\mathcal{R}^g_X = M \setminus X$ *on every spherical space form* $M = \mathbb{S}^n/\Gamma$*, for every closed* X *— an instance of Corollary 3.2.*

(b) *On the round sphere (constant curvature; radius normalised to* 1*), if moreover* $|X| \geq 2$*, then* $M^g_X = M_X$*, and the two-point axis already reconstructs everything:* $\mathcal{R}_X = \mathbb{S}^n \setminus X$*.*

Proof. (a) is Corollary 3.2. (b) Let $a \in M^g_X$ and suppose two distinct minimising geodesics from a reach the same foot x_0. On the round sphere $\exp_a$ is injective on the open ball of radius π, so two distinct geodesics from a of equal length $\leq \pi$ with a common endpoint force that length to be π and x_0 to be the antipode of a. Then $d_X(a) = \pi$ and the free ball $B(a, \pi) = \mathbb{S}^n \setminus \{x_0\}$ avoids X, so $X \subseteq \{x_0\}$ — against $|X| \geq 2$. Hence the minimising geodesics of every $a \in M^g_X$ have at least two distinct feet: $M^g_X \subseteq M_X$, so $M^g_X = M_X$, and (a) gives $\mathcal{R}_X = \mathcal{R}^g_X = \mathbb{S}^n \setminus X$. □

Statement (b) is a genuine constant-curvature rigidity — two minimising geodesics to a common foot force an antipode, so a cut point always presents a second foot — and it fails under variable positive curvature (Example 5.6); on a space form $\mathbb{S}^n/\Gamma$ with $\Gamma \neq \{1\}$ cut points occur below the diameter and only the two-geodesic statement (a) is claimed.

Example 5.5 (A single point on $\mathbb{S}^n$: the antipodal centre)*.* $X = \{x_0\} \subseteq \mathbb{S}^n$. Then $M_X = \varnothing$, so the two-point theory reconstructs nothing. But the antipode $\bar{x}_0$ is a medial centre in M^g_X: it is joined to x_0 by an $(n-1)$-sphere of minimising geodesics, all of length $\pi = d_X(\bar{x}_0)$, and the free ball $B(\bar{x}_0, \pi) = \mathbb{S}^n \setminus \{x_0\}$ swallows everything. Corollary 5.4 thus reconstructs the whole complement from a single point — the extreme demonstration that M^g_X, not M_X, is the reconstruction axis. The capture curve from any $p \neq x_0, \bar{x}_0$ ascends the meridian away from x_0 and reaches $\bar{x}_0$ in time $\pi - d_X(p)$: capture, class (1).

Example 5.6 (Cut-shielding and the two-point axis)*.* Under *variable* positive curvature the two axes genuinely separate: a capturing centre a may have two minimising geodesics to a *single* foot (a cut point of a occurring before any second foot appears), so $a \in M^g_X \setminus M_X$, and a point captured by a is reconstructed by $\mathcal{R}^g_X$ but possibly by no two-point ball. This is the *cut-shielding* obstruction, and it is why the two-point theorem needs a spread hypothesis (Corollary 3.3) or constant curvature (Corollary 5.4(b)), while the two-geodesic theorem needs nothing. The obstruction is quantitative: on prolate ellipsoids of revolution, numerical experiment locates a threshold aspect ratio (polar to equatorial) $c^* = 2.6 \pm 0.1$ past which shielding switches on for a two-point contact set; shielded points are exactly class-(1) points of $\mathcal{R}^g_X$ whose capturing centres lie in $M^g_X \setminus M_X$. An analytic determination of the threshold is open (§9(3)).

5.3 Hyperbolic space and Hadamard manifolds

Let M be Hadamard (complete, simply connected, variable sec ≤ 0). The ideal medial limit functions are Busemann functions at a level (Remarks 2.5 and 4.7), so Theorem 4.3 reads: $\mathcal{R}^g_X$ is the union of the finite medial balls and the shrunken horoballs $\{b_\xi < -c(\xi)\}$ over the defective ideal points ξ. This is the exact Hadamard analogue of Corollary 5.1, with "half-space" replaced by "horoball".

The *faithful* question — which ideal points ξ are defective, and at what level — is the curved analogue of Białożyt's defective-hyperplane certificate. The natural answer mirrors the flat one, with the *horoconvex hull* in the role of the convex hull — for $K \subseteq \partial\mathcal{H}$, $\mathrm{hconv}(K)$ is the trace on the horosphere of the intersection of the horoballs of M containing K, which on a flat horosphere reduces to the ordinary convex hull: ξ is defective iff the contact set $X \cap \partial\mathcal{H}$ on the maximal free horosphere at ξ fails to be horoconvex, $X \cap \partial\mathcal{H} \neq \mathrm{hconv}(X \cap \partial\mathcal{H})$, with level $c(\xi) = 0$. Establishing this certificate is the horoconvex counterpart of Motzkin's theorem and is beyond the flat certificate assumed here; we treat it as part of the faithful-layer programme (§9(1)–(2)) and use, in the examples, only the unconditional statements proved above.

Example 5.7 (Two ideal directions in $\mathbb{H}^2$). Let $X = \{x_0, x_1\}$ in the hyperbolic plane. The bisector is a complete geodesic with two ideal endpoints $\xi_\pm \in \partial_\infty\mathbb{H}^2$. Escaping bisector centres $a_k \to \xi_\pm$ have reaches $r_k = d(a_k, x_0) \to \infty$, and $\psi_{a_k} = d(\cdot, a_k) - r_k \to b_{\xi_\pm}$, a Busemann function at level 0; by the swallow lemma the two open horoballs $\{b_{\xi_\pm} < 0\}$ are reconstructed. Together with the finite bisector balls, which fill a lens-shaped region around the geodesic segment $[x_0, x_1]$, these exhaust $\mathcal{R}^g_X$: every medial centre lies on the bisector, so by Theorem 4.3 every reconstructed point lies in a finite bisector ball or in one of the two horoballs. This is the hyperbolic mirror of Example 5.2, with "open half-plane" replaced by "open horoball" — the escaping centres now run to the two *ideal points* of the bisector rather than off to Euclidean infinity, and the swallowed regions are horoballs there. The contact set on the maximal free horosphere at each $\xi_\pm$ is the single pair $\{x_0, x_1\}$, non-horoconvex — the horoconvex analogue of the defective supporting line. How much of the complement the lens and the two horoballs leave uncovered — the hyperbolic counterpart of the two collinear outer rays — depends on the horoball geometry; horoballs in $\mathbb{H}^2$ are far thinner than half-planes, and the uncovered escape class (3) is correspondingly larger than its flat counterpart (see also Remark 7.7(b)). The swallow and the characterisation above are, in any case, unconditional.

Example 5.8 (The ideal limit is genuinely needed). In both Example 5.2 and Example 5.7 the drifted region is reconstructed by no ball findable from within: a point p deep in the open half-plane / horoball, at an obtuse angle to the bisector, has its capture curve escaping with a single foot (Remark 4.6), so it is class (2), reconstructed only by remote bisector balls, with the ideal limit as the uniform certificate for the whole region. This is why Theorem 4.3 must include the ideal members and why the pure capture theorem (Corollary 3.2) cannot extend to non-compact manifolds.

5.4 Non-compact positive curvature

Here reconstruction is genuinely partial and neither the flat nor the Hadamard certificate applies; the trichotomy still governs.

Example 5.9 (Paraboloid of revolution)*.* Let M be a paraboloid of revolution ($\sec > 0$, a single flattening end) and X a small circle about the vertex. The vertex is a finite medial centre reconstructing the inner disk: class (1). Every point on the flaring side ascends its meridian forever with a single foot, and no medial limit function reaches it: class (3). Hence $\mathcal{R}_X^g \subsetneq M \setminus X$ — partial reconstruction, decided by the non-compactness of M, not by the sign of the curvature.

Example 5.10 (Ideal centres in positive curvature)*.* Take $X = \{x_0, x_1\}$ on the paraboloid. The bisector meridian is unbounded, and its escaping centres furnish genuinely *ideal* medial limit functions on a $\sec > 0$ manifold. There is no contradiction with the boundedness phenomena of positive curvature: it is the maximal free *convex* regions that are bounded under $\sec > 0$, and the sublevel sets $\{\psi < 0\}$ are unbounded *non-convex* free regions. The example shows the ideal class is not a nonpositive-curvature artefact.

5.5 The flat cylinder: a faithful certificate beyond Euclidean space

The simplest manifold outside every certificate above is the flat cylinder $C = (\mathbb{R}/L\mathbb{Z}) \times \mathbb{R}$: flat but not simply connected, so Theorem 1.1 does not apply and neither does the Hadamard picture, and non-compact, so Corollary 3.2 is silent about its ends. Its horofunction boundary consists of two points — the compact circle factor washes out of every limit $d(\cdot, a_k) - d(o, a_k)$ — so ideal medial limit functions are $\pm z + \text{const}$, attached to the two ends; write z for the second coordinate and π for the projection to the circle factor.

Proposition 5.11 (End certificate on the flat cylinder)**.** *Let $X \subseteq C$ be closed, $\varnothing \neq X \neq C$, bounded above: $z_+ := \sup_X z < \infty$. The upper contact set $X_+ := X \cap \{z = z_+\}$ is nonempty and compact, and exactly one of the following holds.*

(i) (Defective end.) *X_+ omits a circle position: $\pi(X_+) \neq \mathbb{R}/L\mathbb{Z}$. Then M_X^g meets every slice $\{z = s\}$ for all large s, the free-ball functions of such centres converge locally uniformly (no subsequence needed) to the medial limit function $\psi = z_+ - z$, and*

$$\{z > z_+\} \;\subseteq\; \mathcal{R}_X^g :$$

the maximal free end-region is reconstructed in full, at level 0*.*

(ii) (Capped end.) *$X \supseteq (\mathbb{R}/L\mathbb{Z}) \times \{z_+\}$. Then every point above the top circle has a unique minimising geodesic to X (the vertical one), no medial ball meets $\{z > z_+\}$, and $\mathcal{R}_X^g \cap \{z > z_+\} = \varnothing$: the free end-region is not reconstructed at all (in particular no ideal medial limit function $c - z$ exists).*

The lower end is symmetric; an end on whose side X is unbounded admits no ideal medial limit function at all ($\psi \geq 0$ on X would fail).

Proof. X closed and $\{z_+ - 1 \leq z \leq z_+\}$ compact make X_+ nonempty and compact; and with $\Theta_\varepsilon := \pi(X \cap \{z \geq z_+ - \varepsilon\})$ the sets Θ_ε are compact, nested, and $\bigcap_{\varepsilon > 0} \Theta_\varepsilon = \pi(X_+) =: \Theta_+$ (a point $(\alpha, a_\varepsilon) \in X$ with $a_\varepsilon \to z_+$ has limit $(\alpha, z_+) \in X$), so $\Theta_\varepsilon \to \Theta_+$ in the Hausdorff sense as $\varepsilon \downarrow 0$.

Two elementary facts about $q = (\theta, s)$ with $s > z_+$. First, the horizontal displacement of any minimising geodesic from q to a foot is at most $L/2$ in absolute value — re-wrapping the lift by $\mp L$ strictly shortens otherwise — with equality forcing a second minimising geodesic (both wraps). Hence, comparing with any top point,

$$s - z_+ \;\le\; d_X(q) \;\le\; \sqrt{(s - z_+)^2 + L^2/4} \;\le\; s - z_+ + \frac{L^2}{8(s - z_+)}.$$

Second, every foot (α, a) of q then has $s - a \le d_X(q)$, i.e. $a \ge z_+ - \varepsilon(s)$ with $\varepsilon(s) := L^2/(8(s - z_+)) \to 0$: feet of high points are near-top points, and their positions lie in $\Theta_{\varepsilon(s)}$.

(i) Fix α_0 and $\rho > 0$ with the arc of radius ρ about α_0 disjoint from Θ_+, and s_1 with $\Theta_{\varepsilon(s)}$ contained in the $\rho/2$-neighbourhood of Θ_+ for $s \ge s_1$. Suppose a slice $\{z = s\}$ with $s \ge s_1$ contained no point of M_X^g. Then every (θ, s) has a unique minimising geodesic, whose horizontal displacement $\delta(\theta) \in (-L/2, L/2)$ (strict, by uniqueness) is continuous in θ (Lemma 2.2 with single values), and the foot position is $\theta + \delta(\theta) \bmod L$. As θ increases through one period, $\theta + \delta(\theta)$ is continuous with total increment L, so by the intermediate value theorem its values cover *every* circle position — including the forbidden arc about α_0: a contradiction. So M_X^g meets every slice $\{z = s\}$, $s \ge s_1$; pick a_s there. For (θ, z) in a fixed compact set and s large, the displacement bound gives $s - z \le d((\theta, z), a_s) \le s - z + L^2/(8(s - z))$, while $s - z_+ \le d_X(a_s) \le s - z_+ + L^2/(8(s - z_+))$; subtracting, $\psi_{a_s} \to z_+ - z$ locally uniformly. Lemma 4.1 swallows the sublevel set: $\{z > z_+\} \subseteq \mathcal{R}_X^g$.

(ii) If the full top circle lies in X then for $q = (\theta, s)$, $s > z_+$, the vertical segment to $(\theta, z_+) \in X$ has length $s - z_+$, while any geodesic to any $x \in X$ has length $\ge s - z(x) \ge s - z_+$, with equality throughout only for that vertical segment: the minimising geodesic is unique, so $M_X^g \cap \{z > z_+\} = \varnothing$. And no medial ball — centred, therefore, at height $\le z_+$ — contains a point q above the circle: the minimising geodesic from the centre to q stays in the open ball and crosses $\{z = z_+\} \subseteq X$, contradicting freeness. So $\mathcal{R}_X^g \cap \{z > z_+\} = \varnothing$; and an ideal limit $c - z$ would swallow $\{z > c\}$, so none exists. □

Remark 5.12 (What the cylinder teaches). The dichotomy is decided by X alone — a faithful certificate, on a manifold covered by neither Theorem 1.1 nor the Hadamard formulation. Two degenerations are forced by the *compact* horosphere. First, Białożyt's "non-convex contact set" relaxes to "proper subset of the horosphere circle": a *single* contact point suffices for the two-geodesic drift, the wrap-around geodesic acting as the second contact. For X a single point this gives $\mathcal{R}_X^g = C \backslash X$ in full — the non-compact counterpart of the $|X| = 1$ clause of Corollary 3.2 — while $M_X = \varnothing$ and $\mathcal{R}_X = \varnothing$ exactly as in $\mathbb{R}^n$: on non-simply-connected manifolds the two axes separate already for singletons. Second, the level is always 0 (a drift centre's reach differs from its height above z_+ by $O(1/s)$): compact horospheres leave no room for the offset phenomenon, consistent with the expectation that $c(\xi) > 0$, if it occurs at all, requires inhomogeneous infinite-volume horospheres (§9(2)). The homotopy content of the singleton example is recorded in Remark 6.6: the reconstruction is complete while $C \setminus X \not\simeq M_X^g$ — the two swallowed escape rays are exactly the Aubry set of d_X, and they carry the fundamental group, which the medial complex of §7 recovers from the two ideal cells.

5.6 How much curvature, where

Remark 5.13 (The hypotheses, layer by layer). Since the geometries above range over constant, variable, and merely sign-constrained curvature, we record exactly what each layer assumes;

"sec ≤ 0" etc. always means a *sign* condition on in-general-variable curvature, never constancy, unless said otherwise.

- *The dynamical layer* — capture (Theorem 3.1), swallow (Lemma 4.1), characterisation (Theorem 4.3), trichotomy (Theorem 4.4), and total reconstruction on compact M (Corollary 3.2): *no curvature hypothesis at all*. Variable curvature of either or mixed sign is covered.

- *The Hadamard picture* (§5.3): the sign condition sec ≤ 0, with variable curvature expressly allowed, *together with simple connectedness*. The latter is not cosmetic: it is what identifies horofunctions with Busemann functions of rays (Remark 2.5), and on a non-simply-connected sec ≤ 0 manifold even the horoconvex formulation is unavailable — the flat cylinder's end dichotomy (Proposition 5.11) takes its place there. The certificate itself is open in both cases (§9(1)).

- *The knife-edge* (Theorem 1.1): exact flatness *and* simple connectedness, i.e. $M = \mathbb{R}^n$ itself. A flat cylinder has sec $\equiv 0$ but is not covered by Theorem 1.1 verbatim: its ideal certificates attach to the two ends, not to a visual sphere of directions — they are instead the dichotomy of Proposition 5.11.

- *Constancy of curvature* is essential in exactly one result above: the two-point theorem on the round sphere (Corollary 5.4(b)). Its mechanism — two minimising geodesics to a common foot force an antipode, so a cut point always presents a second foot — is a constant-curvature rigidity, and the conclusion genuinely fails under variable sec > 0 (the prolate-ellipsoid shielding of Example 5.6). The two-*geodesic* counterpart needs no curvature at all (Corollary 3.2).

The completed picture. Two independent axes organise the theory: *compactness* decides partial versus total; the *sign of the curvature* decides the flavour of the faithful certificate. Rows: compactness. Columns: curvature sign. Entries: the two-geodesic theory, with the two-point refinement in brackets.

	sec ≤ 0	sec $\equiv 0$	sec > 0
non-compact	partial; hconv defect at level $c(\xi)$ (§5.3)	partial; conv defect (Białożyt [6])	partial (paraboloid) finite + ideal centres
compact	*total* $\mathcal{R}^g_X$ [M_X: shielding]	*total* $\mathcal{R}^g_X$ [M_X: shielding]	*total* $\mathcal{R}^g_X$ [M_X: shielding; $= \mathbb{S}^n$ ok]

The compact row is Corollary 3.2 — unconditional, all curvatures, $|X| \geq 1$. The non-compact row is Theorem 4.3, with the faithful certificates known on the flat column and expected on the Hadamard column, and open elsewhere (§9). The column heads are sign conditions on variable curvature and govern only the detection flavour; the certificates quoted in the cells carry the stronger hypotheses of their statements — simple connectedness in the first column, $M = \mathbb{R}^n$ exactly at the knife-edge, with the flat cylinder carrying its own end certificate (Proposition 5.11) — and constancy of curvature appears only in the bracketed two-point refinement of the $\mathbb{S}^n$ cell (Remark 5.13).

6 The homotopy layer: capture, the Aubry set, and the limits of swallowing

Remark 3.5 separated the capture theorem from the canonical singular gradient flow of Lieutier and Albano–Cannarsa–Nguyen–Sinestrari, whose purpose is homotopy equivalence. This section runs the two together and settles the homotopy question for the completed theory: when the axis carries the homotopy type of the complement, what obstructs it when it does not, and what either answer has to do with the reconstruction $\mathcal{R}^g_X$. Throughout, Φ denotes the *canonical* flow: the unique generalised-gradient trajectory of d_X through each point of $M \setminus X$ — Lieutier's flow in $\mathbb{R}^n$ [20]; existence and uniqueness [2, Thm. 3.4, Cor. 3.6] in the Euclidean case and [2, §4] on Riemannian domains. The construction is local, its only input being the local semiconcavity of d_X on $M \setminus X$ [1, 21], so it defines Φ for every closed X in every complete M; uniqueness gives the semigroup property. Canonical trajectories satisfy the inclusion (1), so Theorem 3.1 applies to them verbatim; and they are global in time: d_X is nondecreasing along them, so they stay in $M \setminus X$, and speed ≤ 1 together with completeness of M rules out finite-time breakdown. We import two further facts, both local, from that literature.

(F1) *Joint continuity:* $(t, p) \mapsto \Phi_t(p)$ is continuous on $[0, \infty) \times (M \setminus X)$. (Local semiconcavity makes the ascending field one-sidedly Lipschitz, and Grönwall gives locally Lipschitz dependence on the initial point; this is the continuity that underlies the homotopy statements of [20, 2].)

(F2) *Propagation:* Σ is positively invariant under Φ: once a canonical trajectory is singular it stays singular [2, Thms. 3.7 and 4.5]. (Stated there for domains, under curvature bounds; the proof — semiconcavity plus a differential inequality — is local, and a finite time window of a trajectory lives in a compact subset of $M \setminus X$, where the bounds hold. So it applies verbatim here.)

Lemma 6.1 (Capture is permanent; the cut locus is invariant)**.** *Let $p \in M \setminus X$ and suppose the canonical trajectory of p is captured, with first-approach time T as in Theorem 3.1*(i)*. Then*

$$\Phi_t(p) \;\in\; \Sigma \qquad \textit{for every } t > T:$$

the trajectory does not merely approach the axis; it enters it and never leaves. Moreover the cut locus

$$\mathrm{Cut}(X) \;:=\; \overline{\Sigma} \cap (M \setminus X) \qquad (\textit{closure in } M)$$

is positively invariant as well.

Proof. If $\Phi_T \in \Sigma$, apply (F2) from time T. If $\Phi_T \notin \Sigma$, then by the definition of T every interval $(T, T + \delta]$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$ (proof of Theorem 3.1); (F2) from time s_δ gives $\Phi_t \in \Sigma$ for all $t \geq s_\delta$, and $\delta \downarrow 0$ covers all of (T, ∞). For the cut locus: $q \in \mathrm{Cut}(X)$ is a limit of $a_j \in \Sigma$, so by (F1) and (F2), $\Phi_t(q) = \lim_j \Phi_t(a_j) \in \overline{\Sigma}$; and $\Phi_t(q) \in M \setminus X$ always. □

Theorem 6.2 (Uniform capture is a homotopy equivalence)**.** *Let M be complete, X closed, and let $U \subseteq M \setminus X$ be open and positively invariant under the canonical flow, on which capture is uniform: for some $D < \infty$ the trajectory of every $p \in U$ meets Σ by time D. Then $\Phi_D(U) \subseteq \Sigma \cap U$, and the inclusions*

$$\Sigma \cap U \;\hookrightarrow\; \mathrm{Cut}(X) \cap U \;\hookrightarrow\; U$$

are homotopy equivalences, with Φ_D as homotopy inverse and the flow itself as the homotopy.

Proof. $\Phi_D(p) \in \Sigma$ for $p \in U$ by Lemma 6.1 (the trajectory meets Σ at some $t^* \le D$ and stays), and $\Phi_D(p) \in U$ by invariance. Set $r := \Phi_D \colon U \to \Sigma \cap U$, write $i \colon \Sigma \cap U \hookrightarrow U$ for the inclusion, and let $H(p,s) := \Phi_{sD}(p)$, jointly continuous by (F1). Then H is a homotopy in U from id_U to $i \circ r$; and H restricted to $(\Sigma \cap U) \times [0,1]$ takes values in $\Sigma \cap U$ ((F2) and invariance of U), so it is also a homotopy from $\mathrm{id}_{\Sigma\cap U}$ to $r \circ i$. Hence i is a homotopy equivalence. The same two sentences with $\mathrm{Cut}(X) \cap U$ in place of $\Sigma \cap U$ — using the invariance of the cut locus from Lemma 6.1 and $\Phi_D(U) \subseteq \Sigma \cap U \subseteq \mathrm{Cut}(X) \cap U$ — show that $\mathrm{Cut}(X) \cap U \hookrightarrow U$ is a homotopy equivalence, and the inner inclusion follows by two-out-of-three. $\square$

Corollary 6.3 (Bounded distance: topological faithfulness)**.** *Let M be complete, X closed, $\varnothing \ne X \ne M$, and suppose $D := \sup_{M\setminus X} d_X < \infty$. Then escape is impossible, $\mathcal{R}^g_X = M \setminus X$, every component Ω of $M \setminus X$ meets Σ, and*

$$\Sigma \cap \Omega \;\hookrightarrow\; \mathrm{Cut}(X) \cap \Omega \;\hookrightarrow\; \Omega$$

are homotopy equivalences. In particular:

- (a) *on every compact M: $M \setminus X \;\simeq\; M^g_X \;\simeq\;$ $\mathrm{Cut}(X)$, unconditionally — the closed-set counterpart of the classical retraction of $M \setminus \{x_0\}$ onto* $\mathrm{Cut}(x_0)$*;*
- (b) *for every coarsely dense closed set in any complete M — e.g. a periodic $X \subseteq \mathbb{R}^n$ — the complement, an* unbounded *domain outside the scope of [20, 2], has the homotopy type of its two-geodesic axis and of its cut locus.*

With Corollary 3.3 this completes a pairing: $\sup d_X < \mathrm{inj}(M)$ *makes the two-point reconstruction metrically complete ($\mathcal{R}_X = M \setminus X$), and* $\sup d_X < \infty$ *alone makes the two-geodesic picture topologically faithful.*

Proof. Escape forces $d_X(\Phi_t) = d_X(p) + t \to \infty$ (Theorem 3.1(ii)), impossible under the bound; so every $p \notin M^g_X$ is captured, hence $p \in \mathcal{R}^g_X$ (Theorem 3.1(i)), and $M^g_X \subseteq \mathcal{R}^g_X$ trivially: $\mathcal{R}^g_X = M \setminus X$. If a canonical trajectory avoided Σ on $[0,\tau]$, the rate-one identity (2) would give $d_X(p) + \tau \le D$; hence every trajectory meets Σ by time $D - d_X(p) \le D$, and the meeting point lies in the component of its start. Apply Theorem 6.2 with $U = \Omega$ (open; invariant, since trajectories are continuous and stay in $M \setminus X$). Compact M has $D \le \mathrm{diam}(M)$; a δ-dense X has $D \le \delta$. $\square$

Beyond the bound, escape occurs and the statement genuinely changes. The right language is the *Aubry set* of weak KAM theory, and the translation is exact.

Lemma 6.4 (The Aubry set of d_X is the escape class)**.** *Following [10, Def. 1.4], let*

$$\mathcal{A}^*(X) \;:=\; \big\{q \in M \setminus X : \; q = \gamma(t_0) \text{ for some } t_0 > 0 \\ \text{and some arclength } \gamma : [0,\infty) \to M \text{ with } d_X(\gamma(t)) = t \;\; \forall t \ge 0\big\},$$

the union of the infinite calibrated rays of X. Then:

- (i) *$\mathcal{A}^*(X) \cap \Sigma = \varnothing$;*
- (ii) *$\mathcal{A}^*(X)$ is closed in $M \setminus X$;*

(iii) *a point* $p \in M \setminus X$ *lies in* $\mathcal{A}^*(X)$ *if and only if its capture curves escape — that is,* $\mathcal{A}^*(X)$ *is exactly classes* $(2) \cup (3)$ *of Theorem 4.4, and*

$$M \setminus (X \cup \mathcal{A}^*(X)) \;=\; \{\textit{captured points}\} \;\supseteq\; \Sigma,$$

an open set.

Proof. (i) Let $q = \gamma(t_0)$ lie on a calibrated ray. If q had a second minimising geodesic $\sigma \neq \gamma|_{[0,t_0]}$, the curve σ followed by $\gamma|_{[t_0,t]}$ would realise $d_X(\gamma(t)) = t$ for $t > t_0$ with a corner at q, hence could be strictly shortened — contradiction. (The same argument shows that past a point of Σ no minimising geodesic keeps minimising, so calibrated rays avoid Σ outright.)

(ii) Let $q_j = \gamma_j(t_j) \to q \in M \setminus X$ with γ_j calibrated. Then $t_j = d_X(q_j) \to d_X(q) > 0$, and the feet $\gamma_j(0)$ eventually lie in the compact set $\bar{B}(q, 2d_X(q)) \cap X$. Calibrated rays are unit-speed minimising geodesics on every initial segment ([10, Rem. 1.5]); by properness of M a subsequence converges locally uniformly to an arclength ray γ, and $d_X(\gamma(t)) = t$ passes to the limit: $q = \gamma(d_X(q)) \in \mathcal{A}^*(X)$.

(iii) All capture curves from $p \notin \Sigma$ agree until their first approach to Σ: off Σ the inclusion (1) is the ODE of the single unit field, and the trajectory is the arclength extension of the minimising geodesic of p (§2). If that trajectory escapes, it never meets Σ and the rate-one identity holds forever; prepending the minimising segment from the foot of p produces a curve γ with $d_X(\gamma(t)) = t$ for all $t \geq 0$: an infinite calibrated ray through p, so $p \in \mathcal{A}^*(X)$. Conversely let $p = \gamma(t_0) \in \mathcal{A}^*(X)$; by (i) applied at each $\gamma(s)$, $s > 0$, the ray avoids Σ, and since the trajectory off Σ is the extension of the unique minimising geodesic, the trajectory of p is $t \mapsto \gamma(t_0 + t)$: it never meets Σ, and $d_X \to \infty$ along it — escape. Points of Σ are captured at time 0 and excluded from $\mathcal{A}^*(X)$ by (i); openness of the captured set is (ii). □

Theorem 6.5 (Cannarsa–Cheng–Fathi [10, Thm. 1.6], in the vocabulary of the trichotomy)**.** *For every closed* $X \neq \varnothing$ *in every complete* M*, the inclusion*

$$M_X^g \;\hookrightarrow\; M \setminus (X \cup \mathcal{A}^*(X)) \;=\; \{\textit{captured points}\}$$

is a homotopy equivalence: the captured part of the complement always has the homotopy type of the axis *— no bound on* d_X*, no compactness. (Under uniform capture this is Theorem 6.2 with* $U = M \setminus X$*; in general the uniform clock fails and the statement is exactly [10, Thm. 1.6], translated by Lemma 6.4. Their Theorem 1.1 adds that* M_X^g *is always locally contractible.)*

Remark 6.6 (Homotopy follows capture, not swallowing)*.* The three classes of Theorem 4.4 now have distinct topological meanings. Class (1) carries the homotopy type of M_X^g (Theorem 6.5); the Aubry set $\mathcal{A}^*(X) = (2) \cup (3)$ is what the axis cannot see; and the swallow splits $\mathcal{A}^*(X)$ into its reconstructed part (2) and its free part (3) — a *metric* split that the homotopy type ignores. Both failure modes occur.

- *Escape unswallowed (class (3) active).* X the round circle of radius R in $\mathbb{R}^3$: every axis point has the whole circle as feet, and M_X^g is exactly the axis, contractible. The captured set is the open solid cylinder $\{\rho < R\}$ (each foot-ray there crosses the axis), which deformation-retracts to M_X^g in accordance with Theorem 6.5; the Aubry set is the closed shell $\{\rho \geq R\} \setminus X$, and its unswallowed part — the flange $\{z = 0,\ \rho > R\}$, the plane of the circle beyond it — is exactly what $\mathcal{R}_X^g = \mathbb{R}^3 \setminus (X \cup \{z = 0, \rho \geq R\})$ misses. The homotopy type $\mathbb{R}^3 \setminus X \simeq S^1 \vee S^2$ is created across that shell and is invisible to the contractible axis.

- *Escape swallowed (class (2) only) — complete reconstruction without homotopy equivalence.* The flat cylinder with X a single point (Proposition 5.11, Remark 5.12): $\mathcal{R}^g_X = C \setminus X$ is complete, yet $C \setminus X \simeq S^1 \vee S^1$ while M^g_X is the opposite generator, contractible. The dictionary locates the leak exactly: $\mathcal{A}^*(X)$ is the punctured generator through the point — two vertical calibrated rays, entirely of class (2), swallowed by the wrap-around balls, whose non-contractibility (radius beyond the injectivity radius) is what lets them reconstruct without transporting homotopy type. The captured set is the open strip $C \setminus \{\text{generator}\}$, contractible, in accordance with Theorem 6.5. Completeness of the reconstruction is a metric statement; it does not imply topological faithfulness.

Under $\sup d_X < \infty$ the Aubry set is empty — d_X is unbounded along a calibrated ray — and the two completions re-fuse: that is Corollary 6.3. Emptiness of $\mathcal{A}^*(X)$ is strictly weaker than bounded distance: inside a parabola in $\mathbb{R}^2$ every normal crosses the symmetry axis before it stops minimising, so $\mathcal{A}^*(X)$ misses the interior component entirely and the (unbounded, unboundedly deep) interior is homotopy equivalent to its medial ray, both contractible. Section 7 closes the leak: what the axis cannot see, the ideal medial limit functions can — every swallowed-uncaptured point lies in an ideal free horoball (Theorem 7.3), and gluing those cells onto the axis restores the homotopy type of $\mathcal{R}^g_X$ (Theorem 7.5). The leak itself is not a curvature phenomenon: it recurs for three points in the plane, where complete reconstruction and a contractible axis coexist with a wedge of three circles (Proposition 7.8).

Two classical instances, and the smooth dual side. The corollary and proposition that close the section read Corollary 6.3 back into classical geometry: uniform capture *is* the spine theory of cusped hyperbolic and crystallographic geometry; and for smooth X the homotopy equivalence is the boundary of a mapping-cylinder structure whose inward dual is the Thom space.

Corollary 6.7 (Spines: cusped hyperbolic manifolds and crystallographic quotients)**.** (a) *Let M be a connected complete hyperbolic manifold of finite volume with cusps, and let X be the union of a system of disjoint closed horoball neighbourhoods of the cusps. Then* $\sup d_X < \infty$ *and*

$$M \;\simeq\; M \setminus X \;\simeq\; M^g_X \;\simeq\; \mathrm{Cut}(X),$$

where $\mathrm{Cut}(X)$ *is the Ford–Voronoi spine of the system — the complex dual to the Epstein–Penner decomposition attached to the same horoballs [15]. The spine theorem of cusped hyperbolic geometry is uniform capture for the horoball contact set.*

(b) *Let $M = \mathbb{R}^n/\Gamma$ be a compact flat manifold, $x_0 \in \mathbb{R}^n$, and $X = \{\pi(x_0)\}$ (or any finite set). Then* $\mathrm{Cut}(X)$ *is the projected wall complex of the Voronoi diagram of the orbit* Γx_0*, and $M \setminus X \simeq M^g_X \simeq$* $\mathrm{Cut}(X)$*. For the square torus and a single point,* $\mathrm{Cut}(X)$ *is the union of the two "antipodal" circles — a figure-eight, matching $T^2 \setminus \{x_0\} \simeq S^1 \vee S^1$ — while $M_X = \varnothing$: the singleton has no two-point axis at all, and the two-geodesic axis alone carries the type. The homotopy type that the flat cylinder's complete reconstruction fails to transmit (Remark 6.6) is carried honestly the moment the manifold is compact.*

Proof. (a) $M \setminus X$ is the open thick-with-tubes part, of compact closure (Margulis; finite volume), so $\sup d_X < \infty$ and Corollary 6.3 applies. For the identification of the axis, lift to the universal cover $\mathbb{H}^n$: the preimage of X is a Γ-invariant family of disjoint closed horoballs. A horoball is convex, so a point outside it has a unique nearest point on it, joined by a unique minimising geodesic; hence two minimising geodesics to X end on two *distinct* horoballs of the lifted family, and Σ is the wall set of the Voronoi decomposition of $\mathbb{H}^n$ determined by the family — whose quotient closure is, by definition, the Ford–Voronoi spine. (The two feet may well project to the *same* point of X in M, so it is M_X^g, not M_X, that carries the spine — the two-geodesic axis is again the right object.) Finally $M \simeq M \setminus X$: the closed cusp neighbourhoods retract onto their horospherical boundaries, and the closed thick part onto the open one along a collar. (b) One level down: the lift of X is the orbit Γx_0 (a Delone set; finitely many orbits for a finite set), Σ projects from the walls of its classical Voronoi diagram, and d_X is bounded by the covering radius, so Corollary 6.3 applies. On $\mathbb{R}^2/\mathbb{Z}^2$ with $x_0 = 0$: the Voronoi cell is the unit square centred at 0, its boundary projects to the two circles $\{x = \frac{1}{2}\} \cup \{y = \frac{1}{2}\}$, meeting only at $(\frac{1}{2}, \frac{1}{2})$: a wedge of two circles. ☐

Proposition 6.8 (Smooth contact set: the captured part is a mapping cylinder)**.** *Let X be a closed embedded C^∞ submanifold of the complete manifold M, $S(\nu)$ its unit normal bundle,*

$$\rho(v) \;=\; \sup\{t > 0 : \; d_X(\exp(tv)) = t\} \;\in\; (0, \infty]$$

the cut time, $S_{\text{fin}} = \{\rho < \infty\}$, and $\sigma : S_{\text{fin}} \to M \setminus X$, $\sigma(v) = \exp(\rho(v)v)$, the swallow map*, whose image* $\mathrm{Cu}(X) := \sigma(S_{\text{fin}})$ *is the classical cut locus. Then $\mathcal{A}^*(X) = \{\exp(tv) : \rho(v) = \infty, \; t > 0\}$, and the normal exponential descends to a homeomorphism*

$$\mathrm{Cyl}(\sigma) \;:=\; \big(S_{\text{fin}} \times (0,1] \;\sqcup\; \mathrm{Cu}(X)\big)\big/\big[(v,1) \sim \sigma(v)\big] \;\xrightarrow{\;\approx\;}\; M \setminus \big(X \cup \mathcal{A}^*(X)\big),$$
$$(v,t) \mapsto \exp\!\big(t\rho(v)\, v\big) :$$

for smooth X, with no further hypothesis, the captured part of the complement is the mapping cylinder of the swallow map*, and it strong deformation retracts onto* $\mathrm{Cu}(X)$*. In particular* $\Sigma \subseteq \mathrm{Cut}(X) \subseteq \mathrm{Cu}(X)$*, the inclusion* $\Sigma \hookrightarrow \mathrm{Cu}(X)$ *is a homotopy equivalence for every smooth closed X (Theorem 6.5 and two-out-of-three) — the equivalence of [10] sharpened, on the smooth case, to a retraction onto the full cut locus — and* $\mathrm{Cut}(X) = \mathrm{Cu}(X)$ *when* $\dim X = 0$*, by Bishop's theorem [7]. If $\mathcal{A}^*(X) = \varnothing$ — equivalently $\rho < \infty$ everywhere, in particular whenever* $\sup d_X < \infty$ *— then $M \setminus X$ itself is the cylinder, and under uniform capture the whole chain* $\Sigma \hookrightarrow \mathrm{Cut}(X) \hookrightarrow \mathrm{Cu}(X) \hookrightarrow M \setminus X$ *consists of homotopy equivalences.*

Proof. Minimising geodesics from a point to X arrive orthogonally (first variation), so every $p \in M \setminus X$ is $\exp(d_X(p)v)$ for some $v \in S(\nu)$ with $d_X(p) \le \rho(v)$; the calibration interval $\{t : d_X(\exp(tv)) = t\}$ is $(0, \rho(v)]$ (closed at ρ by continuity); and $\rho(v) = \infty$ makes $t \mapsto \exp(tv)$ an infinite calibrated ray, so Lemma 6.4 identifies $\mathcal{A}^*(X)$ with the union of the open rays $\{\exp(tv) : t > 0\}$, $\rho(v) = \infty$. Positivity of ρ is the tubular neighbourhood theorem; continuity of $\rho : S(\nu) \to (0, \infty]$ is the classical continuity of the cut time, proved for submanifolds as for points (see [4] and the references there, and [25, III.4] for a point).

Write $F(v,t) = \exp(t\rho(v)v)$ on $S_{\text{fin}} \times (0,1]$, continuous since ρ is finite and continuous on the open set S_{fin}. *Image and surjectivity:* a captured p outside Σ has a unique foot direction v, and $\rho(v) = \infty$ would put $p \in \mathcal{A}^*(X)$; so $p = F(v, d_X(p)/\rho(v))$. A point $p \in \Sigma$ has each of its directions v stopped at p — past a point of Σ no minimising geodesic keeps minimising (Lemma 6.4(i)) — so $\rho(v) = d_X(p)$ and $p = \sigma(v)$: thus $\Sigma \subseteq \mathrm{Cu}(X)$ and F maps onto the captured part; and the image avoids $\mathcal{A}^*(X)$, since a point of $\mathcal{A}^*(X) \cap \operatorname{im} F$ would carry two distinct minimising directions and lie in Σ, while $\mathcal{A}^*(X) \cap \Sigma = \varnothing$. *Injectivity off $t = 1$:* if $p = F(v,t)$ with $t < 1$ then $d_X(p) < \rho(v)$; a second representation would again put $p \in \Sigma$ and force $d_X(p) = \rho$, a contradiction — the same argument shows $F(v,t) \notin \mathrm{Cu}(X)$ for $t < 1$. Hence F descends to a continuous bijection $\bar{F} : \mathrm{Cyl}(\sigma) \to M \setminus (X \cup \mathcal{A}^*(X))$. *Properness:* let K be a compact subset of the captured part, $0 < \varepsilon \le d_X \le C$ on K. Feet of points of K lie in the compact $X_K = \{x \in X : \mathrm{dist}(x, K) \le C\}$; if $v_k \in S_{\mathrm{fin}}|_{X_K}$ contribute to $\bar{F}^{-1}(K)$ and $v_k \to v$ with $\rho(v) = \infty$, then the limit point $\lim F(v_k, t_k) = \exp(sv)$ with $s = \lim t_k \rho(v_k) \in [\varepsilon, C]$ would lie in $\mathcal{A}^*(X) \cap K$ — impossible; so the directions stay in a compact subset of S_{fin} on which $\rho \le R$, the F-preimage of K is a closed subset of that compactum $\times [\varepsilon/R, 1]$, and $\mathrm{Cu}(X) \cap K$ is compact because $\mathrm{Cu}(X)$ is closed in the captured part (if $\sigma(v_k) \to q$ captured, feet confined give $v_k \to v$ after a subsequence, $\rho(v_k) = d_X(\sigma(v_k)) \to d_X(q)$ finite, so by continuity $\rho(v) = d_X(q)$ and $q = \sigma(v)$). A proper continuous bijection onto a locally compact Hausdorff space is a homeomorphism; the cylinder coordinate $H_s(v,t) = (v,\, t + s(1-t))$, the identity on $\mathrm{Cu}(X)$, is a strong deformation retraction. Finally $\mathrm{Cut}(X) \subseteq \mathrm{Cu}(X)$: if $a_k \in \Sigma \to q \in M \setminus X$, pick feet x_k of a_k with normal directions $w_k \in S(\nu)$; points of Σ sit at their own cut time (the corner argument above), so $\rho(w_k) = d_X(a_k) \to d_X(q)$; the feet are confined, so after a subsequence $w_k \to w$, and continuity of ρ gives $\rho(w) = d_X(q) < \infty$ and $q = \sigma(w) \in \mathrm{Cu}(X)$. (In particular $\overline{\Sigma}$ avoids $\mathcal{A}^*(X)$ for smooth X — a statement we do not claim for general closed X, where no continuous cut time is available: the merging of directions that keeps M_X^g from being closed is not, by itself, excluded from happening along a calibrated ray.) For $\dim X = 0$, Bishop's theorem [7] applied at the strictly nearest point gives $\mathrm{Cu}(X) \subseteq \overline{\Sigma}$, whence Cut = Cu. The final chain is Theorem 6.2 plus the retraction just built plus two-out-of-three. □

Remark 6.9 (The two flows, and the Thom space). Proposition 6.8 is the outward half of a picture whose inward half is due to Basu and Prasad [4] (see also the survey [24]): for a closed embedded submanifold X of a complete M, the square d_X^2 is Morse–Bott on $M \setminus \mathrm{Cu}(X)$, its gradient flow deformation-retracts $M \setminus \mathrm{Cu}(X)$ onto X, and for closed M the collapse $M/\mathrm{Cu}(X)$ is the Thom space of the normal bundle of X. The same normal geodesics thus carry three retractions: inward to X below the cut time [4]; outward to $\mathrm{Cu}(X)$ across it (Proposition 6.8); and — the only one that survives when X is an arbitrary closed set — the canonical flow into the axis, which turns at Σ instead of stopping (Theorem 6.2). For smooth X the three compute the homotopy type of every space in sight from medial data: $M \setminus (X \cup \mathcal{A}^*) \simeq \mathrm{Cu}(X)$, $M \setminus \mathrm{Cu}(X) \simeq X$, $M/\mathrm{Cu}(X) \cong \mathrm{Th}(\nu)$. On the flat cylinder with $|X| = 1$ (Remark 6.6) the picture is visible whole: $\rho = \infty$ exactly at the two vertical directions — the Aubry rays — and the captured strip is the mapping cylinder of $\sigma : S^1 \setminus \{\pm\} \to \{\text{opposite generator}\}$. Two further notes. *(i) Stability:* $\mathrm{Cu}(X)$ is Hausdorff-stable under C^2 perturbations of the metric and of the submanifold [5], so the cylinder structure is a robust model, in the spirit of the λ-medial axis [11]. *(ii) Retraction versus equivalence:* for arbitrary closed X, even Lieutier's Euclidean equivalence [20] is, to our knowledge, not known to improve to a deformation retraction — the obstruction is the continuity of a canonical centre selection — so the smooth case, where the retraction is explicit and strong, is genuinely special.

7 The medial complex: the ideal cells close the leak

Remark 6.6 located the failure of topological faithfulness precisely: the axis carries the homotopy type of the captured class (1), and what it cannot see is the Aubry set (2)∪(3), which the swallow splits by a metric criterion the homotopy type ignores. The flat cylinder suggested the repair — glue the free horoballs of the ideal medial limit functions onto the captured part — and the theorem to hope for is that the resulting *medial complex* always carries the homotopy type of

$\mathcal{R}^g_X$. This section proves it, on every complete manifold, for every closed X, with no hypothesis. The entire content is a covering theorem: *the finite layer does not leak.* Every point of class (2) — swallowed, but invisible to its own ascent — lies in an *ideal* free horoball, at depth no smaller than its depth in any finite medial ball. Granting that, the captured set and the ideal free horoballs form an open cover of $\mathcal{R}^g_X$, the homotopy colimit of its diagram is $\mathcal{R}^g_X$ by the gluing calculus of Dugger–Isaksen [14], and the captured vertex is identified with M^g_X by Theorem 6.5.

Throughout, $W := M \setminus (X \cup \mathcal{A}^*(X))$ denotes the captured set, open by Lemma 6.4, and for $p \in M \setminus X$ and $a \in M$ we write

$$u_p(a) \;:=\; d_X(a) - d(a,p),$$

the *margin of* p *at* a: $u_p(a) > 0$ with $a \in M^g_X$ says exactly that the medial ball of a swallows p at depth $u_p(a)$. As a difference of 1-Lipschitz functions, u_p is 2-Lipschitz in a (and in p).

Lemma 7.1 (Margin transport)**.** *Let $x \in M \setminus X$, let Φ be a capture curve from x, and let $T = \sup\{t : \Phi_s \notin \Sigma\ \forall s \le t\}$ be its first approach to the axis as in Theorem 3.1. Then for every $p \in M$ the margin $t \mapsto u_p(\Phi_t)$ is nondecreasing on $[0,T]$ and 2-Lipschitz on all of $[0, T_{\max})$:*

$$u_p(\Phi_t) \;\ge\; u_p(x) \quad (t \le T), \qquad |u_p(\Phi_t) - u_p(\Phi_s)| \;\le\; 2|t-s|.$$

Moreover, on any interval $[0,t]$ on which Φ avoids Σ, the curve is the arclength extension of the minimising geodesic of x: capture curves are unique, and are calibrated geodesics, until their first approach.

Proof. On $[0,T)$ the rate-one identity (2) gives $d_X(\Phi_t) = d_X(x) + t$, while $d(\Phi_t, p) \le d(x,p) + t$ since Φ is 1-Lipschitz; subtracting, $u_p(\Phi_t) \ge u_p(x)$, and continuity extends this to $t = T$. The Lipschitz bound is the 2-Lipschitz property of u_p along a 1-Lipschitz curve. For uniqueness: on an avoidance interval, $d_X(\Phi_t) = d_X(x) + t$ and $d(x, \Phi_t) \le t$ force $d(x,\Phi_t) = t$ (as $d_X(\Phi_t) \le d_X(x) + d(x,\Phi_t)$), so $\Phi|_{[0,t]}$ has length $t = d(x,\Phi_t)$ and is a minimising geodesic; the concatenation [foot of x]$\to x \to \Phi_t$ has length $d_X(x) + t = d_X(\Phi_t)$, hence minimises, hence has no corner at x: $\Phi|_{[0,t]}$ is the geodesic extension of the minimising geodesic of x, as in the proof of Lemma 6.4(iii). $\square$

Lemma 7.2 (Cut rays: grazing trajectories ride the cut locus to infinity)**.** *Let $z \in \mathcal{A}^*(X) \cap \mathrm{Cut}(X)$. Then the (unique) capture curve Φ from z — the shift of the calibrated ray of z, by Lemma 6.4(iii) — satisfies*

$$\Phi_t(z) \;\in\; \mathcal{A}^*(X) \cap \mathrm{Cut}(X), \qquad d_X(\Phi_t(z)) = d_X(z) + t, \qquad \textit{for every } t \ge 0:$$

an escaping ray that touches the closure of the axis stays in it forever.

Proof. By Lemma 6.4(iii) and its proof, every capture curve from z is the arclength shift of the calibrated ray of z; it never meets Σ, and the rate-one identity holds for all time. In particular the *canonical* trajectory of §6 is this ray shift. By Lemma 6.1 the cut locus $\mathrm{Cut}(X) = \overline{\Sigma} \cap (M \setminus X)$ is positively invariant under the canonical flow, so $\Phi_t(z) \in \mathrm{Cut}(X)$; and forward points of a calibrated ray lie in $\mathcal{A}^*(X)$ by definition. $\square$

Whether $\mathcal{A}^*(X) \cap \mathrm{Cut}(X)$ can be nonempty for some closed X — merging medial directions accumulating along an escaping calibrated ray — is the regularity question already flagged in Proposition 6.8 (it is empty for smooth X, by continuity of the cut time; the structure theory of [26, 22] is the natural tool for the general case). The point of Lemma 7.2 is that the theorem below does not need to know: a grazing ray is not an obstruction but a certificate, since it drags medial centres with it to infinity.

Theorem 7.3 (Ideal swallow: the finite layer does not leak)**.** *Let M be complete, X closed, and let $p \in \mathcal{A}^*(X) \cap \mathcal{R}^g_X$ — a swallowed, uncaptured point, class* (2) *of Theorem 4.4. Then for* every *finite witness $a_0 \in M^g_X$ with $m_0 := u_p(a_0) > 0$ there is an* ideal *medial limit function ψ with*

$$\psi(p) \;\leq\; -m_0 \;<\; 0:$$

p lies in an ideal free horoball, at depth at least its depth in any finite medial ball.

Proof. Let γ be the calibrated ray of p, $p = \gamma(t_0)$ with $t_0 = d_X(p)$, and let $\eta : [0, L] \to M$, $L = d(a_0, p)$, be a minimising geodesic from a_0 to p; it stays in the free ball $B(a_0, d_X(a_0)) \subseteq M \setminus X$. Concatenate:

$$\sigma(s) := \eta(s) \ (0 \leq s \leq L), \qquad \sigma(L + t) := \gamma(t_0 + t) \ (t \geq 0).$$

Along η, $d(\eta(s), p) = L - s$ and $d_X(\eta(s)) \geq d_X(a_0) - s$ give $u_p(\sigma(s)) \geq m_0$; along the ray, $d_X(\gamma(t)) = t$ and $d(\gamma(t), p) = t - t_0$ give $u_p = t_0 \geq m_0$ (the second inequality because $u_p(\sigma(\cdot)) \geq m_0$ persists to $s = L$ by continuity, where $u_p(p) = d_X(p) = t_0$). So

$$u_p(\sigma(s)) \;\geq\; m_0 \qquad \text{for all } s \geq 0. \tag{3}$$

The endpoint data are opposite: $\sigma(0) = a_0 \in \Sigma$ is captured (at time 0), while $p = \sigma(L) \in \mathcal{A}^*(X)$; the captured set is open and $\mathcal{A}^*(X)$ is closed in $M \setminus X$ (Lemma 6.4), so, using (3),

$$s^* := \inf\, \sigma^{-1}(\mathcal{A}^*(X)) \;\in\; (0, L], \qquad x_* := \sigma(s^*) \in \mathcal{A}^*(X), \qquad u_p(x_*) \geq m_0,$$

and $\sigma(s)$ is captured for every $s < s^*$.

Case A (the transition grazes the cut locus): suppose there are $s_j \uparrow s^*$ with $\sigma(s_j) \in \overline{\Sigma}$. Then $x_* \in \mathrm{Cut}(X) \cap \mathcal{A}^*(X)$. By Lemma 7.2 the points $w_t := \Phi_t(x_*)$ lie in $\mathrm{Cut}(X)$ with $d_X(w_t) = d_X(x_*) + t \to \infty$, and by Lemma 7.1 (the trajectory never meets Σ, so $T = \infty$) $u_p(w_t) \geq u_p(x_*) \geq m_0$. Pick $a_t \in \Sigma$ with $d(a_t, w_t) \leq \min(1, 1/t)$: then $a_t \in M^g_X$,

$$u_p(a_t) \;\geq\; m_0 - 2/t, \qquad d_X(a_t) \;\geq\; d_X(x_*) + t - 1/t \;\to\; \infty.$$

Since d_X is bounded on bounded sets, (a_t) has no bounded subsequence. By Lemma 4.2 a subsequence of (ψ_{a_t}) converges locally uniformly to a medial limit function ψ with $\psi(p) = \lim(-u_p(a_t)) \leq -m_0$, and ψ is ideal. Done.

Case B (the transition is clean): otherwise there is $s_1 < s^*$ with $\sigma(s) \notin \overline{\Sigma}$ for $s \in [s_1, s^*)$. Take $s_j \uparrow s^*$ in $[s_1, s^*)$ and set $x_j := \sigma(s_j)$: captured, not in $\overline{\Sigma}$. Each x_j has a capture curve Φ^j with finite first approach T_j, and by the dichotomy proof of Theorem 3.1 there are times $t_j \in [T_j, T_j + 1/j]$ with $b_j := \Phi^j_{t_j} \in \Sigma = M^g_X$. By Lemma 7.1,

$$u_p(b_j) \;\geq\; u_p(x_j) - 2/j \;\geq\; m_0 - 2/j, \qquad d_X(b_j) \;\geq\; d_X(x_j) + T_j - 1/j.$$

If (T_j) is unbounded, pass to a subsequence with $T_j \to \infty$: then $d_X(b_j) \to \infty$, and exactly as in Case A, Lemma 4.2 produces an ideal ψ with $\psi(p) \leq -m_0$. If (T_j) is bounded, pass to a subsequence $T_j \to \bar{T} < \infty$. On $[0, T_j)$ each Φ^j avoids Σ, hence (Lemma 7.1) is the geodesic extension of the minimising geodesic of x_j. Since $x_* \notin \Sigma$ ($\mathcal{A}^*(X) \cap \Sigma = \varnothing$), the direction field U is single-valued and continuous at x_* (Lemma 2.2), so the extensions converge locally uniformly to the extension of the minimising geodesic of x_* — which is the calibrated ray of x_* — and, as $d(b_j, \Phi^j_{T_j}) \leq 1/j$,

$$b_j \;\longrightarrow\; z := \Phi_{\bar{T}}(x_*) \;\in\; \mathcal{A}^*(X) \cap \overline{\Sigma}, \qquad u_p(z) = \lim u_p(b_j) \geq m_0:$$

a grazing point with full margin. Case A applied at z (its hypothesis is exactly $z \in \mathcal{A}^*(X) \cap \mathrm{Cut}(X)$) finishes the proof. $\square$

Corollary 7.4 (Ideal decomposition of the reconstruction). *For every complete M and closed X,*

$$\boxed{\mathcal{R}_X^g \;=\; W \;\cup\; \bigcup\big\{\{\psi<0\}:\ \psi \text{ an ideal medial limit function of } X\big\}}$$

— the captured set plus the ideal layer; the finite medial balls are needed only where capture already operates. On the Aubry set the trichotomy is decided by the ideal functions alone:

$$\text{class }(2)=\mathcal{A}^*(X)\cap\bigcup_{\psi\,\text{ideal}}\{\psi<0\},\qquad \text{class }(3)=\mathcal{A}^*(X)\setminus\bigcup_{\psi\,\text{ideal}}\{\psi<0\}:$$

a point escapes unswallowed if and only if it is Aubry and no ideal medial limit function is negative on it. In $\mathbb{R}^n$ this sharpens Theorem 6 of [6]: the half-space layer alone covers the entire drift-swallowed set.

Proof. $W\subseteq\mathcal{R}_X^g$ is Theorem 4.4(1), the horoballs are swallowed by Lemma 4.1, and conversely $\mathcal{R}_X^g\setminus W=\mathcal{A}^*(X)\cap\mathcal{R}_X^g$ is covered by Theorem 7.3. The class identities follow since $\mathcal{A}^*(X)\cap W=\varnothing$. □

Theorem 7.5 (The medial complex carries the reconstruction). *Let M be complete, X closed, and let Ψ be any family of ideal medial limit functions whose free horoballs cover $\mathcal{A}^*(X)\cap\mathcal{R}_X^g$ (by Theorem 7.3 the family of* all *of them qualifies). Then*

$$\mathcal{U}\;:=\;\{W\}\;\cup\;\big\{\{\psi<0\}:\ \psi\in\Psi\big\}$$

is an open cover of $\mathcal{R}_X^g$, and the natural map from the homotopy colimit of its Čech diagram (the members and their finite intersections) to $\mathcal{R}_X^g$ is a weak equivalence [14] — a homotopy equivalence, since these spaces are ANRs. Since $M_X^g\hookrightarrow W$ is a homotopy equivalence (Theorem 6.5), the medial complex *— M_X^g together with the ideal free horoballs, glued along the pattern of their overlaps with the captured set and with one another — carries the homotopy type of $\mathcal{R}_X^g$; and of $M\setminus X$ whenever the reconstruction is complete.*

Proof. Covering is Corollary 7.4 together with $\{\psi<0\}\subseteq\mathcal{R}_X^g$ (Lemma 4.1) and $W\subseteq\mathcal{R}_X^g$; all members are open. The homotopy-colimit statement for an arbitrary open cover is [14]; open subsets of manifolds and their homotopy colimits have CW type, so the weak equivalence upgrades by Whitehead. The last sentence substitutes the CCF equivalence at the captured vertex. □

Corollary 7.6 (Faithfulness without an ideal layer). *If X admits no ideal medial limit function — e.g. whenever M_X^g is bounded, so that no sequence of medial centres escapes — then class* (2) *is empty, $\mathcal{R}_X^g=W$, and*

$$\mathcal{R}_X^g\;\simeq\;M_X^g:$$

the two-geodesic axis carries the homotopy type of everything it reconstructs. This extends the faithfulness regime of Corollary 6.3 beyond bounded distance: d_X may be unbounded and the Aubry set nonempty, as on the paraboloid of Example 5.9 (M_X^g the vertex, $\mathcal{R}_X^g$ the inner disk, both contractible, the entire flaring end escaping), or for the round circle in $\mathbb{R}^2$ (M_X the centre, $\mathcal{R}_X$ the open disk).

Proof. Class (2) $=\mathcal{A}^*(X)\cap\bigcup_{\text{ideal}}\{\psi<0\}=\varnothing$ by Corollary 7.4; so $\mathcal{R}_X^g=W\simeq M_X^g$ by Theorem 6.5. □

The examples, reassembled. On the flat cylinder with X a point (Remark 6.6): W is the open strip, M_X^g the opposite generator, and the two ideal cells are the open half-cylinders $\{\pm z > 0\}$, each a homotopy circle, with $\{z > 0\} \cap \{z < 0\} = \varnothing$ and each W-overlap an open square, contractible. The complex is the double mapping cylinder

$$S^1 \longleftarrow * \longrightarrow * \longleftarrow * \longrightarrow S^1 \quad \simeq \quad S^1 \vee S^1,$$

the figure-eight: exactly $C \setminus X$. The leak of Remark 6.6 is not an anomaly of the theory but the visible size of its ideal cells. For the round circle in $\mathbb{R}^3$: $W = \{\rho < R\}$ is the captured solid cylinder, the two ideal cells are the half-spaces $\{\pm z > 0\}$, all three contractible with contractible overlaps, so the complex is contractible — and indeed $\mathcal{R}_X^g = \mathbb{R}^3 \setminus (X \cup \{z = 0, \rho \geq R\})$ is star-shaped about the centre. Here $M \setminus X \simeq S^1 \vee S^2$ differs from $\mathcal{R}_X^g$: the complex is faithful to the *reconstruction*, and the discrepancy is exactly class (3), the unswallowed flange. For two points in $\mathbb{R}^2$: strip plus two half-planes, all contractible over contractible overlaps; $\mathcal{R}_X^g$, the plane minus two closed rays, is contractible. In all three, finitely many maximal cells suffice as Ψ; Theorem 7.5 is indifferent to the choice.

Remark 7.7 (What the complex decides, and what it does not). (a) *Levels.* Any covering family Ψ computes the same homotopy type, so the complex never needs the optimal level: the level function $c(\xi)$ of Remark 4.7 (item (2) of §9) is untouched, as is the detection question — *which* ideal functions exist is still read from M_X^g, not from X (items (1)–(2)). (b) *The flat no-leak conjecture, reduced — and refuted.* In $\mathbb{R}^n$ every ideal medial limit function is affine and its cell $H = \{\psi < 0\}$ is an open half-space avoiding X; along a calibrated ray $\gamma(t) = x_\gamma + tu$ the affine function $t \mapsto \psi(\gamma(t))$ starts nonnegative ($\psi \geq 0$ on X) and is monotone, so a ray that enters H never leaves: $\mathcal{A}^*(X) \cap H$ is a bundle of inward half-lines. The conjecture that $\mathcal{R}_X = \mathbb{R}^n \setminus X$ forces $\mathbb{R}^n \setminus X \simeq M_X$ is therefore equivalent, by Theorem 7.5, to the statement that these attachments are trivial: $H \cap W \hookrightarrow H$ *and its finite-intersection versions* are weak equivalences. For a single cell both spaces are contractible candidates — H is convex, and $H \cap W$ is H minus the half-line bundle — and on the acute triangle below the single-cell attachments are indeed trivial. The conjecture fails anyway: Proposition 7.8 exhibits double overlaps $H' \cap H''$ that sink entirely into the Aubry set, and three points in the plane already leak all of their homotopy. On a Hadamard manifold the corner mechanism is throttled by the thinness of horoballs — a calibrated ray not asymptotic to the cell's ideal point eventually exits it, so three points in $\mathbb{H}^2$ do not even reconstruct completely — and which complete Hadamard reconstructions leak is open. (c) *Regularity.* When the cover is good — all members and intersections contractible, as below the convexity radius, or for all of $\mathbb{R}^n$ if (b) holds — the nerve theorem [8] replaces the homotopy colimit by a simplicial complex, and Damon's skeletal calculus [13] realises exactly this reconstruction-from-medial-data programme for compact generic regions in $\mathbb{R}^3$; the local-tree and delta-convex structure of [26, 22] is what a CW model of the complex would be built on. (d) *Where the classes live.* The escape flow $E_t(x) := \Phi_t(x)$ restricts to $\mathcal{A}^*(X)$ (unique rays, no branching), is continuous there by Lemma 2.2, and E_R is a homotopy inverse to $\mathcal{A}^*(X) \cap \{d_X \geq R\} \hookrightarrow \mathcal{A}^*(X)$ for every R: classes (2) $\cup$ (3) are carried entirely at infinity, which is why only ideal objects can see them — the geometric content of Theorem 7.3.

Proposition 7.8 (The flat leak: three points refute the no-leak conjecture)**.** *Let $X \subset \mathbb{R}^2$ be three affinely independent points. Then the reconstruction is complete, and*

$$\mathcal{R}_X \;=\; \mathbb{R}^2 \setminus X \;\simeq\; S^1 \vee S^1 \vee S^1, \qquad \textit{while } M_X \textit{ is the Voronoi tripod, contractible.}$$

Complete reconstruction does not transport homotopy type even in $\mathbb{R}^2$, even for $|X| = 3$: the flat no-leak conjecture of Remark 7.7(b) is false.

Proof. Write e for a generic edge of the triangle, ℓ_e for its line, n_e for its outward unit normal and m_e for its midpoint. M_X is the Voronoi skeleton of three points — the circumcentre with its three outward bisector rays — a tripod.

Ideal cells. Along the outward bisector ray the centres $a_t = m_e + tn_e$ have, for all large t, exactly the endpoints of e as feet and $d_X(a_t) = \sqrt{t^2 + h^2}$, $2h = |e|$ (for an obtuse triangle the opposite vertex is the nearer foot for small t on one edge, which changes nothing in the limit); hence $d(x, a_t) - d_X(a_t) \to \psi_e(x) := \langle m_e - x,\, n_e\rangle$ locally uniformly, an *ideal* medial limit function whose cell $H_e = \{\psi_e < 0\}$ is the open half-plane beyond ℓ_e.

The Aubry set. In $\mathbb{R}^n$ the unit-speed ray $t \mapsto p + tv$ from $p \in X$ is calibrated iff $\bigcup_{t>0} B(p + tv, t) = \{x : \langle x - p,\, v\rangle > 0\}$ avoids X, i.e. iff v lies in the normal cone N_p of $\operatorname{conv} X$ at p; and every Aubry point lies on such a ray through its foot (Lemma 6.4). Hence $\mathcal{A}^*(X) = \bigcup_i (p_i + N_{p_i})$, three closed corner cones.

Complete reconstruction. Fix a vertex p, let n', n'' be the normals of its two edges, so $N_p = \operatorname{cone}(n', n'')$ and $p \in \ell' \cap \ell''$. If $v = an' + bn''$ with $a, b \geq 0$ had $\langle v,\, n'\rangle \leq 0$ and $\langle v,\, n''\rangle \leq 0$, then $|v|^2 = a\,\langle n',\, v\rangle + b\,\langle n'',\, v\rangle \leq 0$, so $v = 0$. Thus every $v \in N_p \setminus \{0\}$ has, say, $\langle v,\, n'\rangle > 0$, and since $p \in \ell'$,

$$\psi'(p + tv) \;=\; -t\,\langle v,\, n'\rangle \;<\; 0 \qquad (t > 0):$$

the ray lies in the adjacent cell from the instant it leaves p. So $\mathcal{A}^*(X) \subseteq H_1 \cup H_2 \cup H_3$, and Corollary 7.4 gives $\mathcal{R}_X^g \supseteq W \cup \mathcal{A}^*(X) = \mathbb{R}^2 \setminus X$, i.e. $\mathcal{R}_X = \mathcal{R}_X^g = \mathbb{R}^2 \setminus X$ ($M_X = M_X^g$ in $\mathbb{R}^n$).

Finally $\mathbb{R}^2$ minus three points is a wedge of three circles, and the tripod is contractible. □

Remark 7.9 (Anatomy of the flat leak). (a) *The corner mechanism.* For an acute triangle the failure is isolated exactly where Remark 7.7(b) did not look. Each single-cell attachment is trivial: $W \cap H_e$ is the strip between the two corner cones flanking e, contractible, as is H_e. But two cells adjacent at a vertex p overlap in the nonempty open cone $H' \cap H'' = p + \{v : \langle v,\, n'\rangle > 0,\ \langle v,\, n''\rangle > 0\} \subseteq p + N_p \subseteq \mathcal{A}^*(X)$ (the containment because the normals subtend an angle $\geq 90°$): the double overlap sinks *entirely* into the Aubry set, $W \cap H' \cap H'' = \varnothing$, and the finite-intersection attachment fails as badly as possible. The cover $\{W, H_1, H_2, H_3\}$ is good (all nonempty pieces convex or strips; $H_1 \cap H_2 \cap H_3 = \varnothing$ since the weighted outward normals sum to zero), so the nerve theorem applies: the nerve is the complete graph K_4 — four contractible vertices, six edges, every triangle missing — which is a wedge of three circles. Theorem 7.5 thus computes $\mathcal{R}_X$ exactly; it is $W \hookrightarrow \mathcal{R}_X$ that fails. The three leaked classes are localised by winding: a loop winding once about p must cross the unbounded cone $p + N_p \subseteq \mathcal{A}^*(X)$, so no captured loop winds about any vertex, while the cycle $W \to H' \to H'' \to W$ around p winds once. For an *obtuse* triangle the cover is not even good: the obtuse corner's overlap cone protrudes past its normal cone, the protrusion is captured near the apex and Aubry far out (inside the opposite vertex's cone), and both $W \cap H' \cap H''$ and one $W \cap H_e$ become disconnected — the goodness threshold of item (5b) of §9 is visible already for triangles: the ideal cover of three points is good precisely when no angle is obtuse.

(b) *The leak count for finite sets.* Let $X \subset \mathbb{R}^2$ be finite and affinely spanning, $k = |X|$, with no point of X in the relative interior of a hull edge, and let V be the number of hull vertices. Consecutive hull vertices bound an unbounded Voronoi edge, so every hull edge is defective as above; the corner computation of Proposition 7.8 covers every normal cone, and $\mathcal{R}_X = \mathbb{R}^2 \setminus X \simeq \bigvee_k S^1$. On the other side M_X is the Voronoi skeleton, whose first Betti number is the number of *bounded* Voronoi cells, i.e. of interior points of X. The leak therefore has rank exactly $V \geq 3$: one class per hull corner. Complete reconstruction is generic for finite planar sets; what fails, by exactly the number of Aubry corners, is transport.

(c) *Corners are the mechanism.* The notched half-plane $X = \{y \leq 0,\ |x| \geq 1\}$ has $\mathcal{A}^*(X) = \{y > 0,\ |x| \geq 1\}$, the vertical rays over the shelf, and no hull corner; the single ideal cell $\{y > 0\}$ (centres $(0, t)$, feet $(\pm 1, 0)$) covers the Aubry set at level 0, the reconstruction is complete, and there is no leak: $\mathbb{R}^2 \setminus X$ and $M_X = \{x = 0\}$ are both contractible. Separately, the *single-cell* attachment can itself fail further afield: for $X =$ unit circle $\cup$ the plane $\{z = -10\}$ in $\mathbb{R}^3$, the cell $H = \{z > 0\}$ is defective (centres on the circle–plane bisector bowl escape with $\psi \to -z$), its Aubry trace is the cylinder shell $\{\rho = 1, z > 0\}$, and $H \cap W =$ (open tube) $\sqcup$ (outside) is disconnected — though there the reconstruction is incomplete below the plane.

(d) *The moral, inverted.* The leak of Remark 6.6 is not a curvature phenomenon: it is affine, generic, and quantified by the corners of the convex hull. What the cylinder adds is only that the cells themselves can be homotopy circles; the flat leak builds its loops from contractible cells glued around a corner. Lieutier's theorem is genuinely a theorem about bounded domains — completeness of the reconstruction does not restore it in the unbounded setting — but the medial complex does, everywhere (Theorem 7.5).

8 Discussion: the layers of the theory

It is worth stating plainly what has been achieved and what the shape of the theory is, because the metric and the topological statements are easy to conflate.

The unconditional layer is complete. On every complete Riemannian manifold and for every nonempty closed proper X, the reconstruction $\mathcal{R}^g_X$ is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3); every point is captured, drift-swallowed, or escaping (Theorem 4.4); and reconstruction is total on every compact manifold (Corollary 3.2). No curvature bound, genericity, analyticity, spread, or cardinality hypothesis appears anywhere in this layer. The two ideas that make it work are worth isolating:

- *Stop at the axis, do not flow through it.* The margin one would like to accumulate is not monotone through multi-geodesic points (Remark 3.4); but the axis is the *goal*, so stopping at its first approach both avoids the non-monotonicity and delivers capture with the margin intact. This is what frees the compact theorem from any swallow property past cut points, and from every regularity question at focal and conjugate degenerations.
- *Normalise limits by the reach.* Measuring a free ball by its depth $d_X(a)$ rather than by the distance to a fixed basepoint makes the limit objects basepoint-free and absorbs the historical "offset" into the function itself (Remarks 2.5 and 4.7). The swallow lemma then costs three lines on *any* complete manifold.

The faithful layer is curvature-specific and only partly known. Replacing "ψ is a medial limit function" by a certificate readable from X and the geometry of M is exactly the content of Białożyt's theorem in the flat case (Corollary 5.1) and of the end dichotomy on the flat cylinder (Proposition 5.11); it is vacuous on compact manifolds, has a natural horoconvex formulation on Hadamard manifolds (§5.3), and is open in mixed curvature (§9). The honest summary is: the *dynamics* of reconstruction are understood in full generality; the *combinatorics* of which asymptotic directions carry a defect are understood exactly where a Motzkin-type theorem is available, and Białożyt's theorem is the flat instance of such a theorem.

The topological layer cuts across both. Homotopy type is transported by capture and by nothing else: the captured part of the complement always has the homotopy type of the axis (Theorem 6.5), the equivalence is total when $\sup d_X < \infty$ (Corollary 6.3), and swallowing — metrically complete as it may be — transmits nothing (Remark 6.6: the flat cylinder; three points in the plane). The medial complex is the exact repair: the finite layer does not leak (Theorem 7.3), so the axis with the ideal free horoballs glued on carries the homotopy type of $\mathcal{R}_X^g$ unconditionally (Theorem 7.5). Note the asymmetry of the two failure classes: class (3) is invisible to $\mathcal{R}_X^g$ but perfectly visible to the ideal layer, being exactly the Aubry points on which every ideal medial limit function is nonnegative (Corollary 7.4); class (2) is reconstructed by $\mathcal{R}_X^g$ but its homotopy is carried only by the complex. And the leak is affine, not Riemannian — the corner cones of a convex hull already produce it (Proposition 7.8) — so the complex is not a tax paid for curvature: it is the correct object even in $\mathbb{R}^n$, where Lieutier's bounded-domain theorem does not extend to unbounded complements.

9 What remains

The theory is now unconditional in its dynamical form on every complete Riemannian manifold, and, with §7, in its homotopy form as well. What remains open is exactly the *faithful* layer — certificates decidable from X and the geometry of M, with M_X^g absent — together with the regularity of the medial complex; items (1)–(2) and (5) live on the non-compact (escape) side, and (3)–(4) are their compact-frontier companions.

(1) *Certificates in mixed curvature.* For which ideal points (horofunctions) h of a general non-compact M is $h + c$ a medial limit function, read from the contact structure of X on $\{h = 0\}$? Known: conv/Motzkin in $\mathbb{R}^n$ [6, 23], and the flat cylinder, whose end certificate is the sharp dichotomy of Proposition 5.11; vacuous on compact M; formulated, via the horoconvex hull, on Hadamard manifolds (§5.3), where proving the horoconvex counterpart of Motzkin's theorem is the first item of business. Unknown already for a complete surface with both signs of curvature; and unwritten for the cusped hyperbolic surface (horocyclic cross-sections shrinking up the cusp) and the general flat quotient $\mathbb{R}^n/\Gamma$ (toral and mixed ends), which should follow the cylinder's two steps — horofunction ends, then a wrap-around forcing argument — but are outside the theorems as stated.

(2) *The level function.* On an inhomogeneous Hadamard manifold, characterise $c(\xi) = \inf\{c : b_\xi + c \text{ medial limit}\}$ from X alone (Remark 4.7); the level vanishes in every instance computed here, and no example with $c(\xi) > 0$ is known — deciding whether the deep-part phenomenon of Remark 4.7 actually occurs is the sharpest open question on the Hadamard side.

(3) *The two-point axis in variable curvature.* Quantify cut-shielding (Example 5.6): the prolate threshold $c^* = 2.6 \pm 0.1$ awaits an analytic determination, and more generally the class of compact M with $\mathcal{R}_X = M \setminus X$ for every closed X with $|X| \geq 2$ (the round sphere qualifies, Corollary 5.4(b); bounded aspect ratio conjecturally).

(4) *Chebyshev problems on the frontier.* The detection questions the method isolates at its boundary: Motzkin's theorem in Carnot groups ($\mathbb{CH}^n$ horospheres) and in Alexandrov

spaces of indefinite curvature (pinched horospheres); and the domain (Lieutier) problem on $\mathbb{S}^n$, where the boundary's positively curved intrinsic geometry becomes the detection object.

(5) *Regularity of the medial complex.* The homotopy question across the Aubry set is settled by §7: the medial complex — M_X^g with the ideal free horoballs glued on — always carries the homotopy type of $\mathcal{R}_X^g$ (Theorem 7.5), because the finite layer does not leak (Theorem 7.3). What remains is its structure. *(a) The flat leak, localised:* the flat no-leak conjecture is settled, negatively — three points in the plane reconstruct completely ($\mathcal{R}_X = \mathbb{R}^2 \setminus X$, the sets characterised by [6, Thm. 6]) and leak all three of their loops past a contractible axis (Proposition 7.8); for finite planar X the leak has rank exactly the number of hull corners, and it dies without corners (Remark 7.9). What survives is localisation: is the leak in $\mathbb{R}^n$ likewise carried by the non-smooth strata of $\partial \operatorname{conv} X$, and which pairs of homotopy types $(\mathbb{R}^n \setminus X,\ M_X)$ are realised by completely reconstructing sets? *(b) A CW or nerve model:* when is the ideal cover good, so that the nerve theorem [8] computes the complex (for three points: iff the triangle has no obtuse angle, Remark 7.9(a)) — and can the local-tree and delta-convex structure theory of [26, 22], or Damon's skeletal calculus [13], produce a finite model in the generic compact-contact case? *(c) Grazing rays:* can $\mathcal{A}^*(X) \cap \operatorname{Cut}(X) \neq \varnothing$ actually occur for a closed (necessarily non-smooth, Proposition 6.8) contact set? Lemma 7.2 shows the theory is indifferent, but the example, or its impossibility, would calibrate how wild the transition between capture and escape can be.

Conclusion. Białożyt's Theorem 6 generalises to every complete Riemannian manifold, in two layers. The *unconditional* layer is proved here in full: reconstruction by the two-geodesic axis is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3); every point is captured, swallowed, or escaping (Theorem 4.4); capture is total on compact manifolds (Corollary 3.2), with no hypotheses whatsoever beyond completeness and closedness. The *faithful* layer — replacing "is a medial limit" by a certificate on contact sets — is exact on the flat knife-edge (conv, Theorem 1.1), proved on the flat cylinder (the end dichotomy of Proposition 5.11), formulated on Hadamard manifolds (hconv, §5.3), and trivial on compact manifolds; its extension into mixed curvature, together with the Chebyshev-set problems it generates, is the genuine residue. The two analytic obstructions that block the passage from $\mathbb{R}^n$ — the containment step past cut points and the radial offset — are removed, the first by stopping the ascent at its first approach to the axis instead of flowing through it, the second by normalising the limit objects by their own reach. A topological layer sits on top: the captured part of the complement always has the homotopy type of the axis (Theorem 6.5, via the Aubry dictionary of Lemma 6.4), totally so when $\sup d_X < \infty$ (Corollary 6.3); the flat cylinder separates the two completions — swallowing finishes the reconstruction across the Aubry set, but only capture transports homotopy type — and the medial complex re-fuses them: the finite layer does not leak (Theorem 7.3), so the axis with its ideal free horoballs glued on carries the homotopy type of $\mathcal{R}_X^g$ on every complete manifold, for every closed contact set (Theorem 7.5); the leak the complex repairs is already flat — three points in the plane leak all of their homotopy past a contractible axis, one loop per hull corner (Proposition 7.8) — and the residue of the homotopy programme is regularity, not existence (§9(5)). In the smooth case the layer is classical geometry made medial: the captured part is the mapping cylinder of the swallow map onto the cut locus, its collapse is the Thom space [4], and its uniform-capture instances are the Ford–Voronoi and crystallographic spines

(Corollary 6.7, Proposition 6.8).

References

[1] P. Albano and P. Cannarsa, *Structural properties of singularities of semiconcave functions*, Ann. Scuola Norm. Sup. Pisa **28** (1999), 719–740.

[2] P. Albano, P. Cannarsa, K. T. Nguyen and C. Sinestrari, *Singular gradient flow of the distance function and homotopy equivalence*, Math. Ann. **356** (2013), 23–43.

[3] J.-P. Aubin and A. Cellina, *Differential Inclusions*, Grundlehren 264, Springer, 1984. (Existence and extension for upper semicontinuous convex compact right-hand sides, §2.1.)

[4] S. Basu and S. Prasad, *A connection between cut locus, Thom space and Morse–Bott functions*, Algebr. Geom. Topol. **23** (2023), 4185–4233.

[5] A. Bhowmick, *Stability of the injectivity radius and the cut locus of submanifolds under perturbations*, arXiv:2605.26499 (2026).

[6] A. Białożyt, *Sets reconstructable with medial axis*, arXiv:2604.20671 (2026). (§5.2 poses the compact Riemannian reconstruction question settled by Corollary 3.2.)

[7] R. L. Bishop, *Decompositions of cut loci*, Proc. Amer. Math. Soc. **65** (1977), 133–136.

[8] K. Borsuk, *On the imbedding of systems of compacta in simplicial complexes*, Fund. Math. **35** (1948), 217–234.

[9] M. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Grundlehren 319, Springer, 1999. (Horofunctions and Busemann points, II.8.)

[10] P. Cannarsa, W. Cheng and A. Fathi, *Singularities of solutions of time dependent Hamilton–Jacobi equations. Applications to Riemannian geometry*, Publ. Math. Inst. Hautes Études Sci. **133** (2021), 327–366.

[11] F. Chazal and A. Lieutier, *The λ-medial axis*, Graphical Models **67** (2005), 304–331.

[12] J. Cheeger, *Critical points of distance functions and applications to geometry*, in: Geometric Topology: Recent Developments, Lecture Notes in Math. 1504, Springer, 1991, 1–38.

[13] J. Damon, *The global medial structure of regions in $\mathbb{R}^3$*, Geom. Topol. **10** (2006), 2385–2429.

[14] D. Dugger and D. C. Isaksen, *Topological hypercovers and $\mathbb{A}^1$-realizations*, Math. Z. **246** (2004), 667–689.

[15] D. B. A. Epstein and R. C. Penner, *Euclidean decompositions of noncompact hyperbolic manifolds*, J. Differential Geom. **27** (1988), 67–80.

[16] V. Gershkovich and H. Rubinstein, *Morse theory for Min-type functions*, Asian J. Math. **1** (1997), 696–715.

[17] M. Gromov, *Hyperbolic manifolds, groups and actions*, in: Riemann Surfaces and Related Topics, Ann. of Math. Stud. 97, Princeton, 1981, 183–213. (The horofunction compactification.)

[18] K. Grove, *Critical point theory for distance functions*, in: Differential Geometry: Riemannian Geometry, Proc. Sympos. Pure Math. **54**, Part 3, Amer. Math. Soc., 1993, 357–385.

[19] K. Grove and K. Shiohama, *A generalized sphere theorem*, Ann. of Math. **106** (1977), 201–211.

[20] A. Lieutier, *Any open bounded subset of $\mathbb{R}^n$ has the same homotopy type as its medial axis*, Computer-Aided Design **36** (2004), 1029–1046.

[21] C. Mantegazza and A. C. Mennucci, *Hamilton–Jacobi equations and distance functions on Riemannian manifolds*, Appl. Math. Optim. **47** (2003), 1–25. (Semiconcavity and first variation of the distance from a closed set.)

[22] T. Miura and M. Tanaka, *Delta-convex structure of the singular set of distance functions*, Comm. Pure Appl. Math. **77** (2024); arXiv:2204.10449.

[23] T. Motzkin, *Sur quelques propriétés caractéristiques des ensembles bornés non convexes*, Atti Accad. Naz. Lincei **21** (1935), 773–779.

[24] S. Prasad, *Cut locus of submanifolds: a geometric and topological viewpoint*, arXiv:2303.14931 (2023).

[25] T. Sakai, *Riemannian Geometry*, Transl. Math. Monogr. 149, Amer. Math. Soc., 1996.

[26] M. Tanaka and S. V. Sabau, *The cut locus and distance function from a closed subset of a Finsler manifold*, Houston J. Math. **42** (2016), 1157–1197.